A UNIVERSITY OF OREGON AND OREGON STATE UNIVERSITY COLLABORATION

FALL 2025

TIMBER TECTONICS IN THE DIGITAL AGE

LED BY

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INTRODUCTION

Our class proposal for the Tualatin Riverfront park is a kit-of-parts system designed using key connections and laminated plywood.To reinforce the reusability of the structure, students were restricted to the use of only wooden connections; metal fasteners were not allowed. Students also used plywood reclaimed from the 2024 Timber Tectonics studio.We have built one of a set of small portals, designed to be placed at entries and exits to the park. We also propose two sibling structures as a compliment to the Juanita Pohl center and future boat launch, which will provide 2000 square feet of gathering space in the language of the smaller structure.

The community favored a design which was true to the origins of place, celebrating the nearby trestle bridge's structure and Oregon's natural habitats. A new boat launch, ADA accessible parking, and a vehicle access loop were key desires.The proposed structures will revitalize Tualatin's Riverfront Park. The structures will also aid in the effort to connect visitors from the Tualatin Lake at the Commons to the Tualatin Greenway, Juanita Pohl Center, and Van Raden Community Center. The larger proposed spaces, as well as individual portals, will enforce wayfinding features and delineate clear entrances and exits to the park. These structures will include spaces for retreat, venues for entertainment and events, and storage and rental services.

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DESIGN CONCEPT & PROCESS

Class Design Objectives

- To create a community-appropriate structure with echoes to vernacular architecture
- To provide shelters that compliment and connect to nearby park amenities To create a reusable, classic structure which can be easily relocated or removed
- To take advantage the unique opportunity to place a major community gathering space in the outdoor setting of the local park

Design Steps

- Key precedents were found in Tualatin's Trestle bridge, the rustic structure already present on the Tualatin Riverfront, and

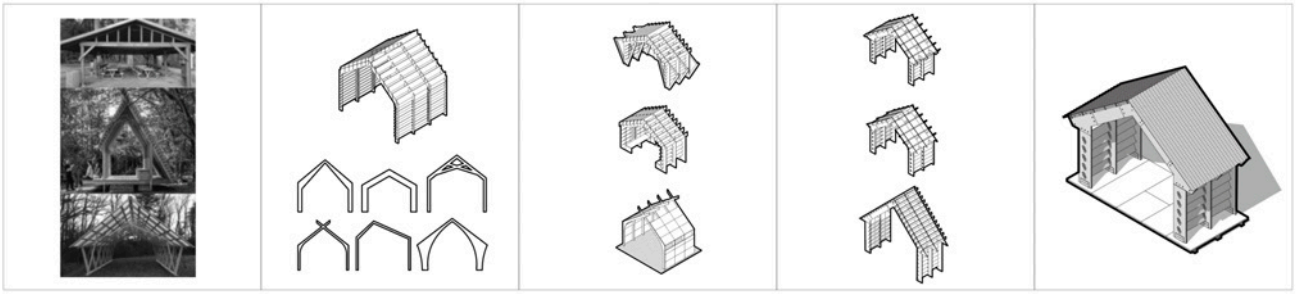
Future Steps

- Based on review, the design should be revised to create a stronger experience of a "transformative" portal
- The kit of parts portal should differ in language from the proposed buildings
- Larger building setback from the curb and increased path size around the boathouse, as well as garage doors or better ventilation on the boathouse, would be improvements

Design Process Precedents

Based on the midterm's encouragement of a classic form which exposed the wood connections at play, we leaned towards a simple structure that would accomplish architectural and enclosure needs at once.

Curved profiles of the portals, as well as an enhanced overhang created by sacrificial 2 by 6 wood pieces, were proposed. Ultimately, material restrictions saw a clean and small portal emerge out of an idea of lateral bracing as a "shingle."



Stage 1: Early Inspiration | *Stage 2: Gable Form* | *Stage 3: Experimentation* | *Stage 4: Refinement* | *Stage 5: Final Result*

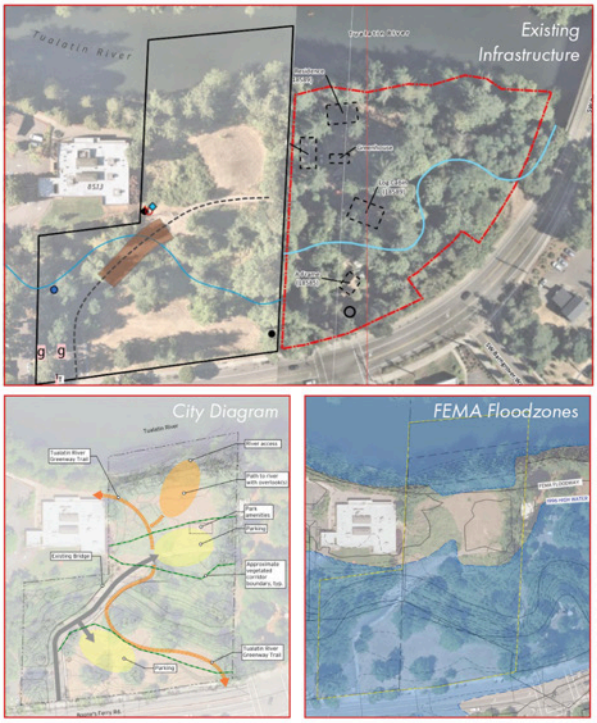
Design Process Diagram

Tualatin Riverfront Park Proposed Site Plan

The chosen Riverfront park site is located in downtown Tualatin, critically adjacent to the Tualatin Lake at the Commons as well as future housing developments in the area and a potential new farmer's market located in the current parking lot south of the park.

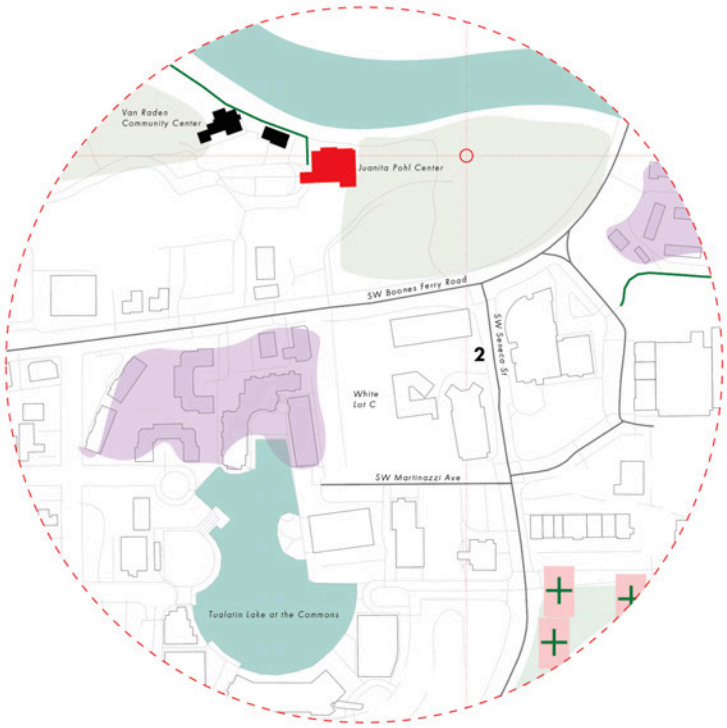
The site has key restrictions in its placement near a riparian zone in need of restoration. A potential pathway under the bridge as well as the extension and new openings of currently dammed creeks introduces new ways for visitors to interact with water.

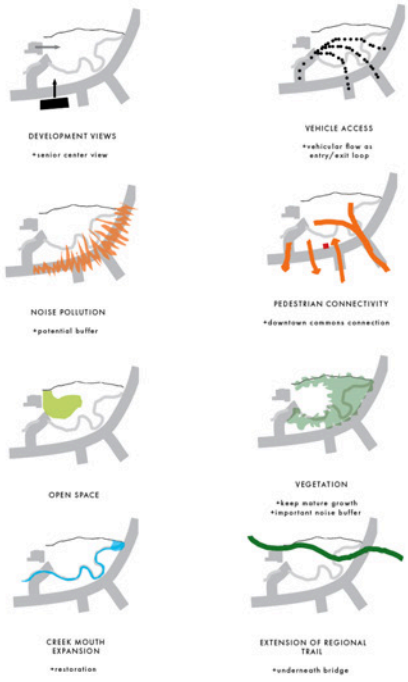
The community specifically wanted all proposed plans to include ADA accessible parking spots and safety from flood zones. A new boat launch for non-motorized boats needed to accommodate the safe transport and storage of kayaks.



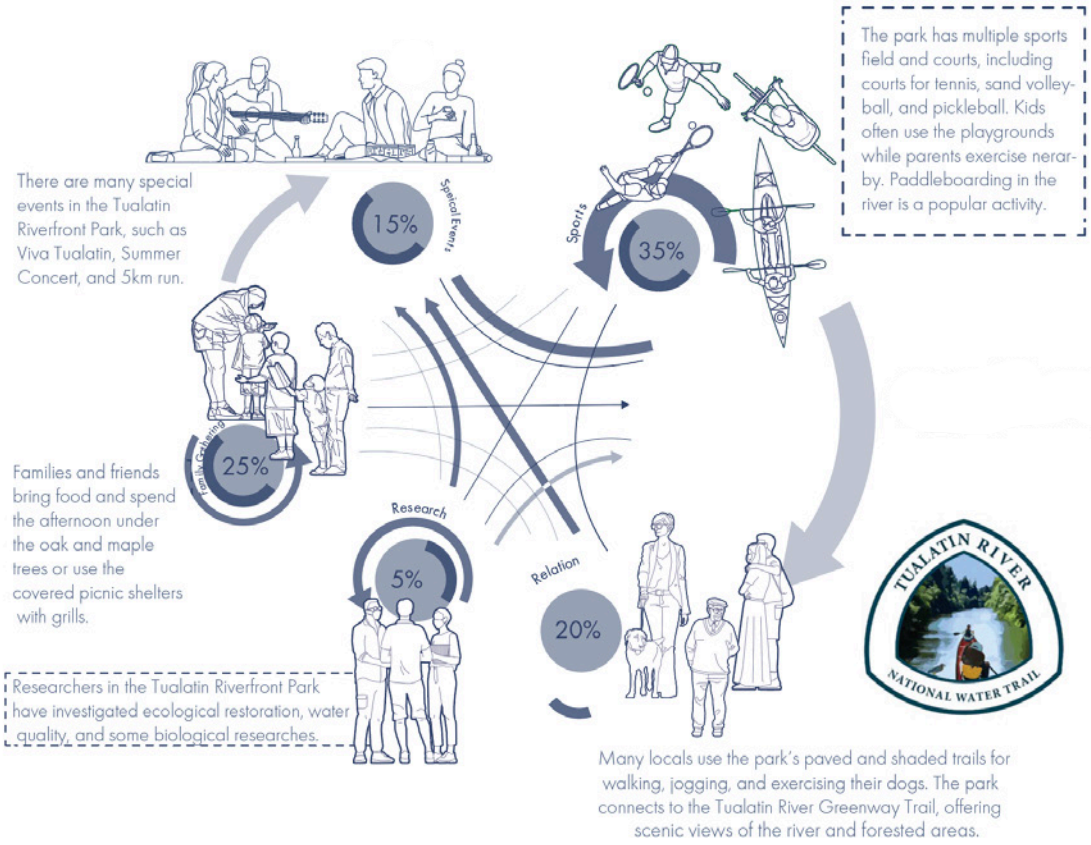
Site Features

Adjacencies

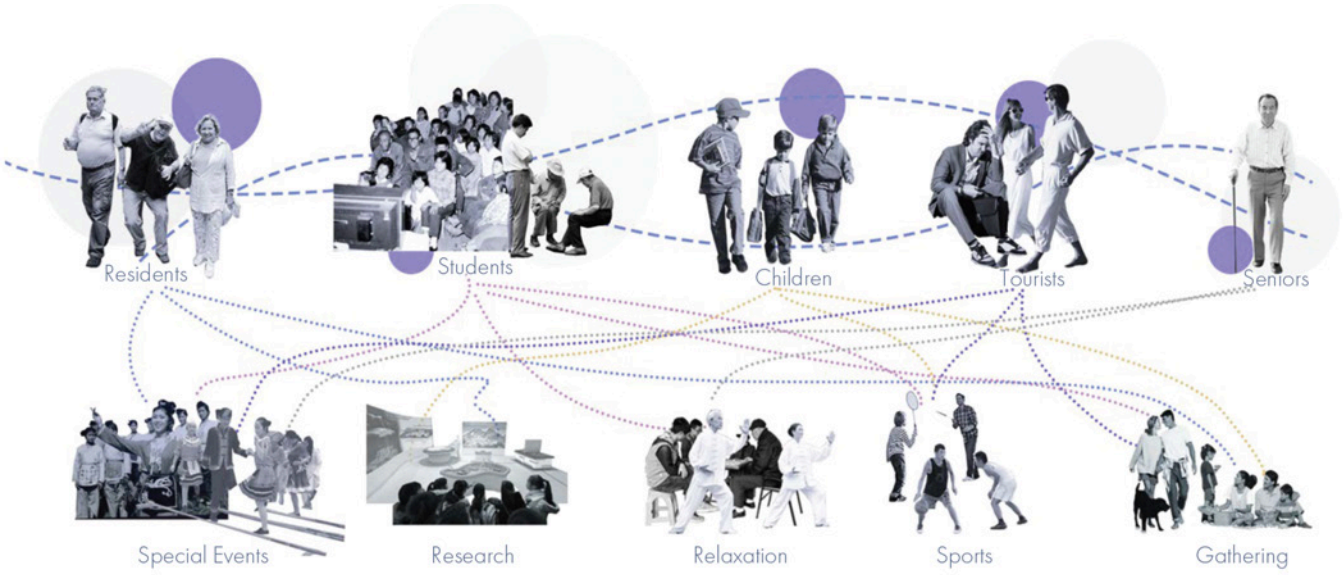




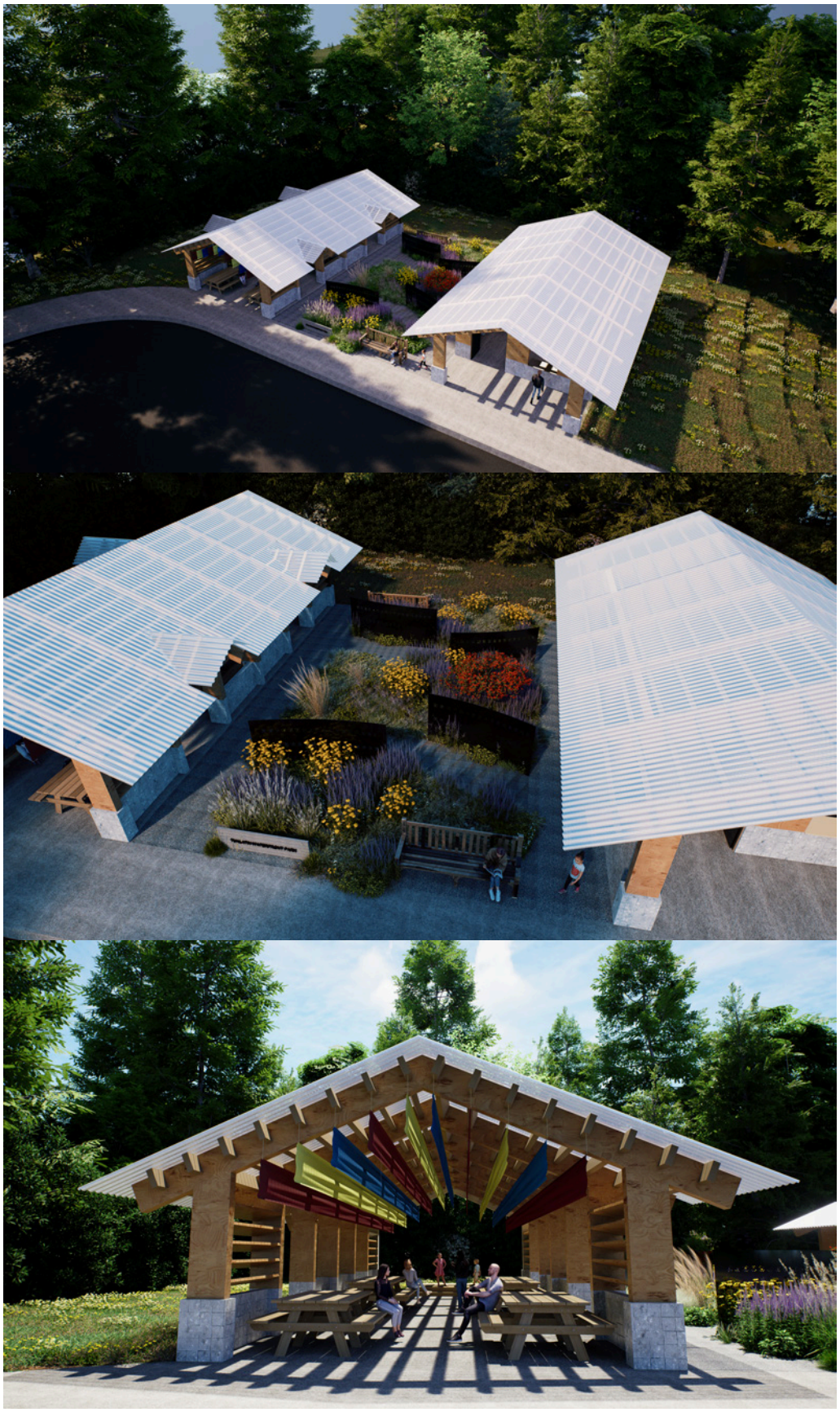
Roads and Views

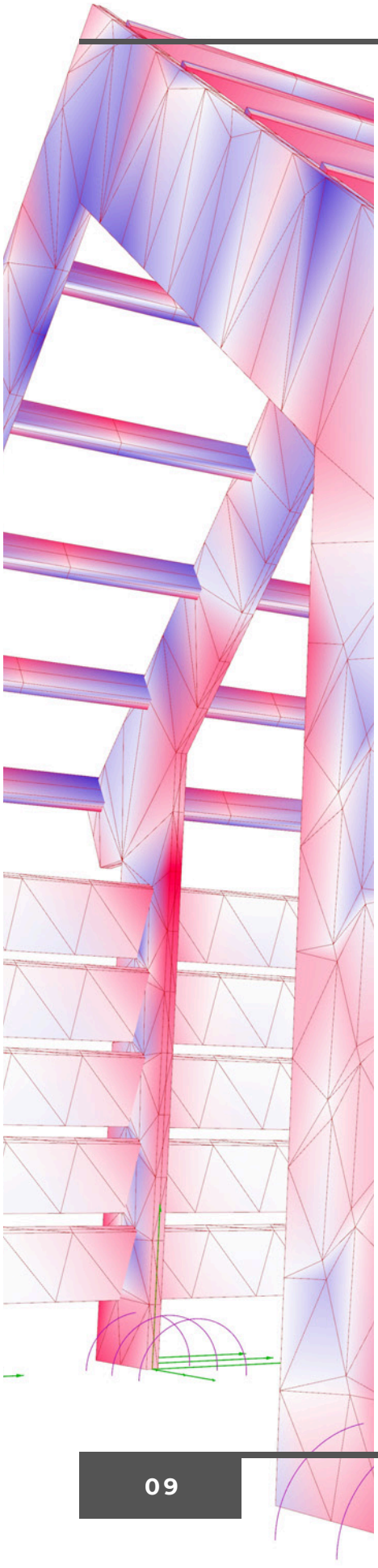


Daily Events



Activity Map





STRUCTURAL DESIGN & ANALYSIS

in our structural system, the laminated plywood frame handles the primary loading and is stabilized by secondary systems made from wood, which are the purlins and angled slats.

The team as a whole faced challenges due to the innovative design of the structural members, which evolved over time.

- It evolved from a dovetail design to dowel connection
- We considered a reciprocal frame before settling on a traditional frame structure
- The reclaimed material faces uncertainties regarding the quality of the member materials
- Lateral bracing design helps to resist some lateral forces

NDS 2024 sets the standards to follow for the dowel-type connections, but not very much for wooden dowels. Most dowel types used in timber are metal. We chose to implement a dowel size of 1-inch diameter, based on several methods used in larger scale timber connections. Consequently, a 3-inch minimum spacing, both from the element edge and center-to-center between dowels, was incorporated, according to NDS provisions. Our team adopted an unconventional and innovative design in the overall structure, due to lack of research with wooden dowel connections, as discussed previously.

The metal base connections were recommended by researchers at the Tallwood Design Institute, and after various tests for both structural integrity and constructability, we elected to use two (2) steel angle brackets per column.

KARAMBA 3D ANALYSIS

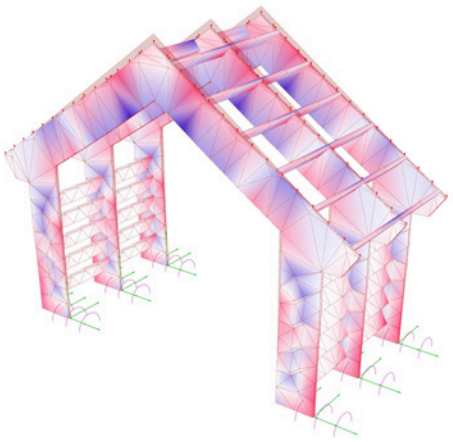
Karamba 3D allowed the ability to simulate and analyze simple 3D models of various design iterations throughout the term. We were able to accurately define material properties and dimensions in order to analyze load applications in different scenarios throughout the design process.

The analyses from Karamba 3D throughout the design process output displacement values which then influenced critical design choices like member depth, connection placement, and more structural roofing system considerations for the final build.

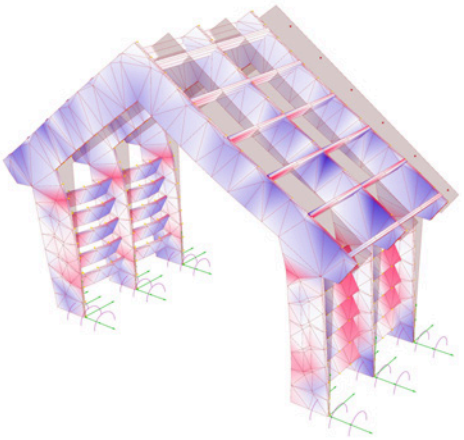
The Karamba analysis was able to prove that structural stability is achievable amongst a roofing system dead load and potential wind loads for the final design proposal. The bracing panels between the frames were also proven to provide enough lateral stability within the frame even with a smaller cross section of 1/2".

Potential modifications and adjustments to the structure:

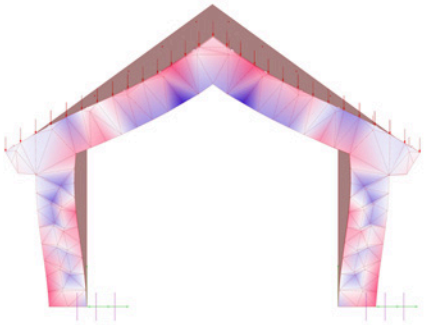
- Modular roofing system following rest of structure
- Thicker members, columns and panels
- Ground connection thrust force counter design
- More even dowel spacing, for better distribution of loads.



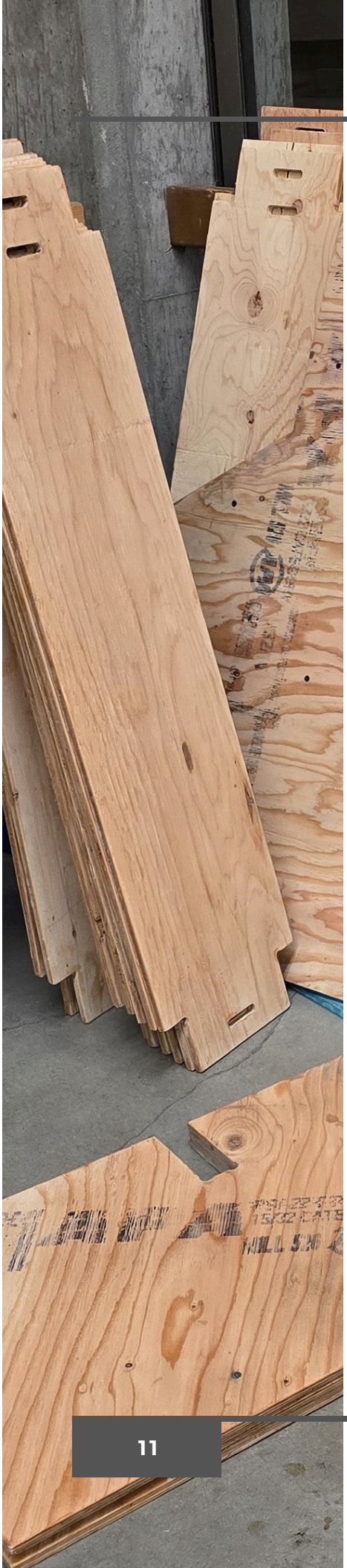
Karamba 3D Model



Deflection from force or wind load perpendicular to frame depth



Deformation from a much heavier roof system or snow load



MATERIALS

For this project, both the UO and OSU instructors pushed for two key restraints in our design: We had to focus on mainly using wood-to-wood connections, unless otherwise approved, and the materials had to be recycled wood from the previous project made by our counterparts from last year

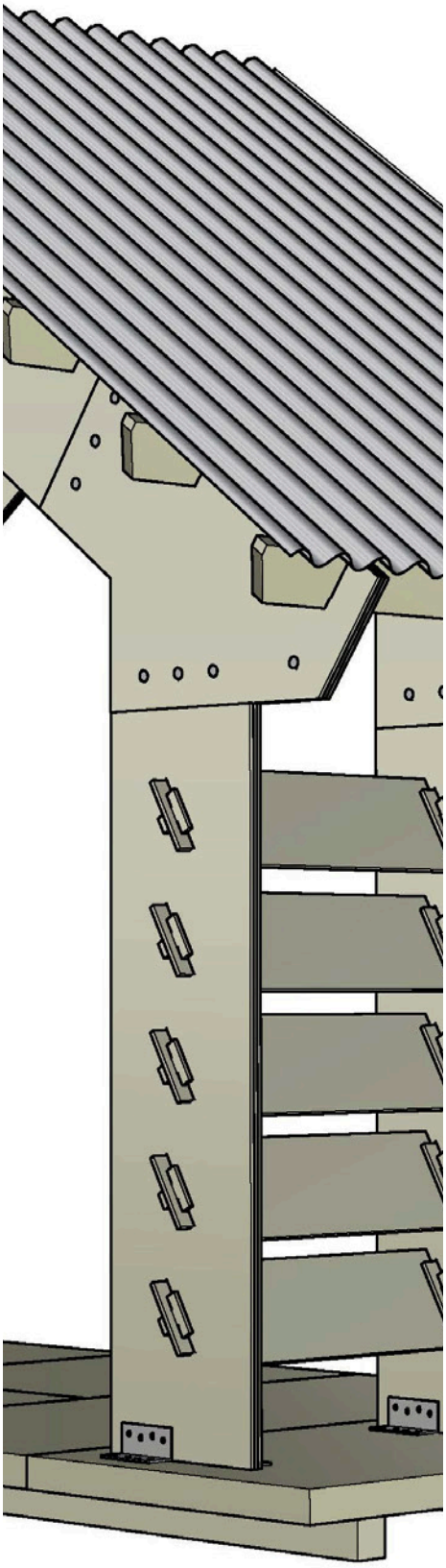
Our team had planned to use reclaimed material consisting of what was originally 1/2-inch plywood, but testing concluded that it not sturdy enough for the conceptual designs at the time. Opportunities were found in laminating three layers of the 1/2-inch material together, creating a thicker material as well as customizing pieces for a tongue-and-groove system for our dowel connections and 2x6 boards.

Our approach for re-laminating and CNC machining the plywood frame elements has similarities to the process for Mass Plywood Panels (MPP), and we collaborated with the Tallwood Design Institute, using a high strength wood glue, a wood lamination press, and CNC cuts for the designs in the Emmerson Lab at Oregon State University.

The material quality was somewhat lowered due to the fact that our material was reclaimed wood from past projects. Reused wood faces several issues that could affect the design, and so layering the plywood gave the best structural strength we could achieve without wasting materials.

TALLWOOD

DESIGN INSTITUTE



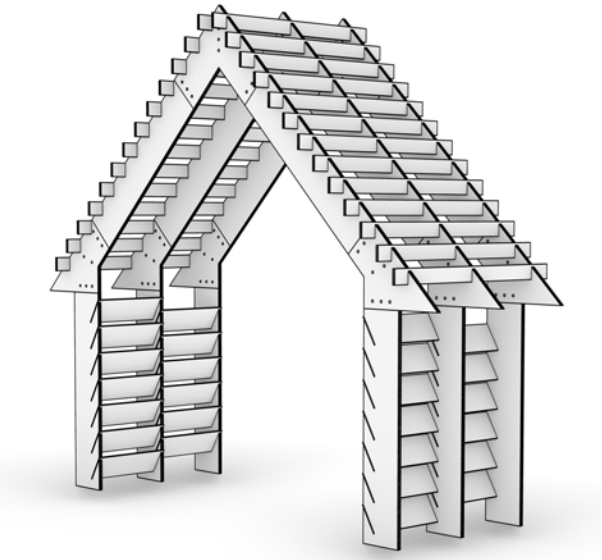
KIT OF PARTS

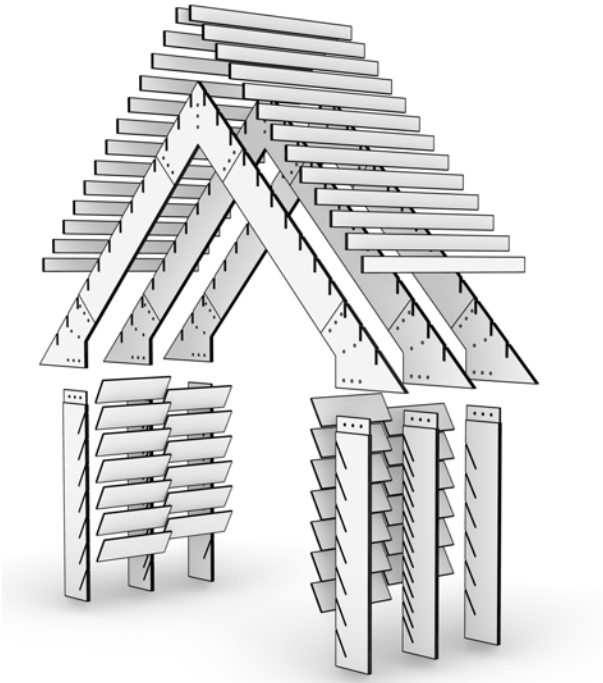
This kit of parts was developed based on the idea of a frame system with lateral stiffening panels seen in some of our precedents from both Veneer House and Wiki House. We aimed to create a modular system that could repeat and elongate with the addition of repeating frames and spanning panels with unique end conditions on either end so as to elegantly finish the system.

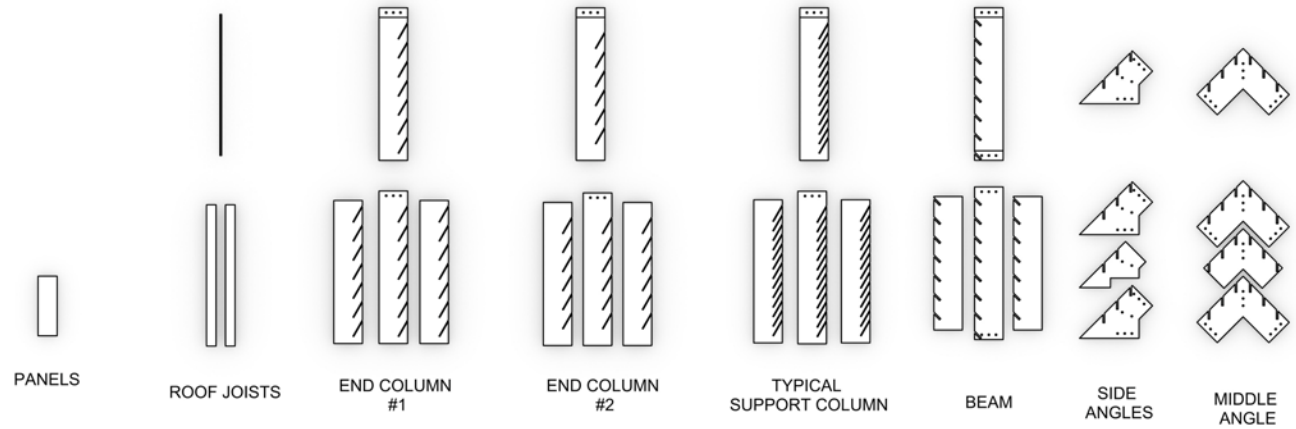
The design of this system was iterative mostly in the expression of the various connection types and the proportions of the system rather than in the aesthetic design. However there was some iteration in regards to the edge conditions of the frame corners and the look of the panels from the elevations. We wanted the system to be expressive of its construction, but in an elegant way.

INITIAL KIT OF PARTS

Elements of this first proposal below stayed consisted in our design throughout the design process, including the dowel connections within the frame and the frame itself as the fundamental basis of design. In this first iteration however the slats would just slide in and out of the columns which was inevitably changed due to worries about public use, lateral stability, and stiffness.

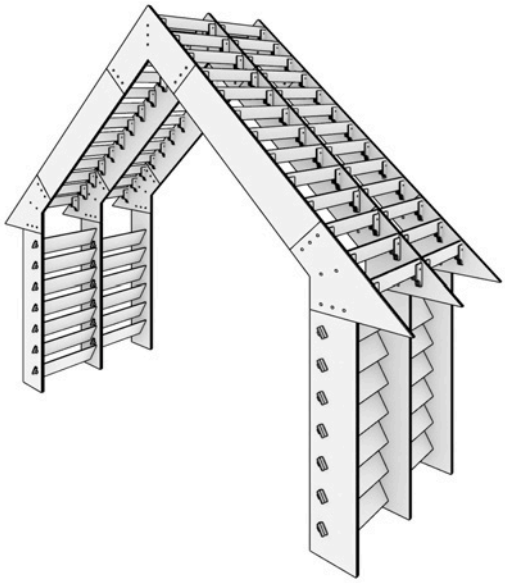


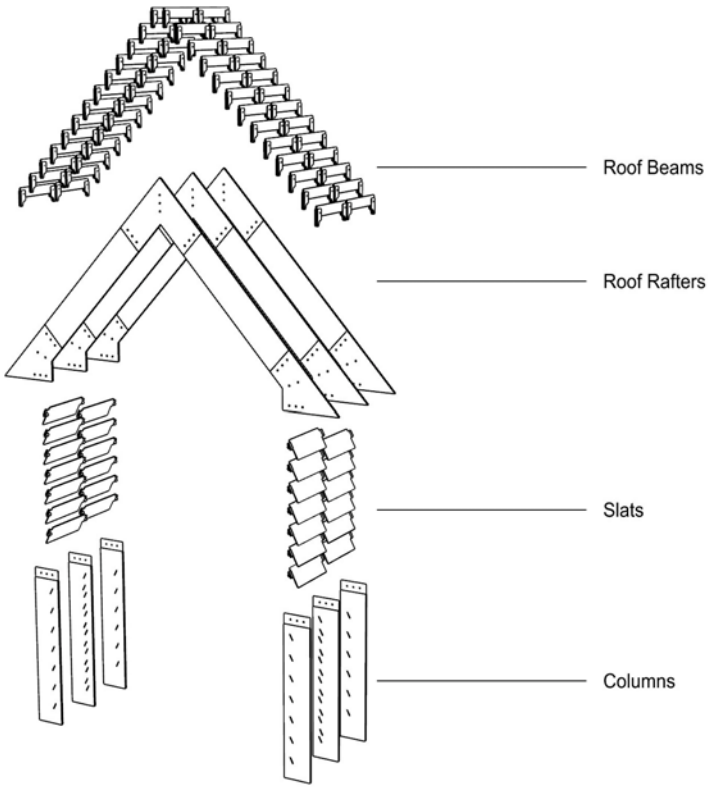




CONSTRUCTABILITY REVIEW
KIT OF PARTS

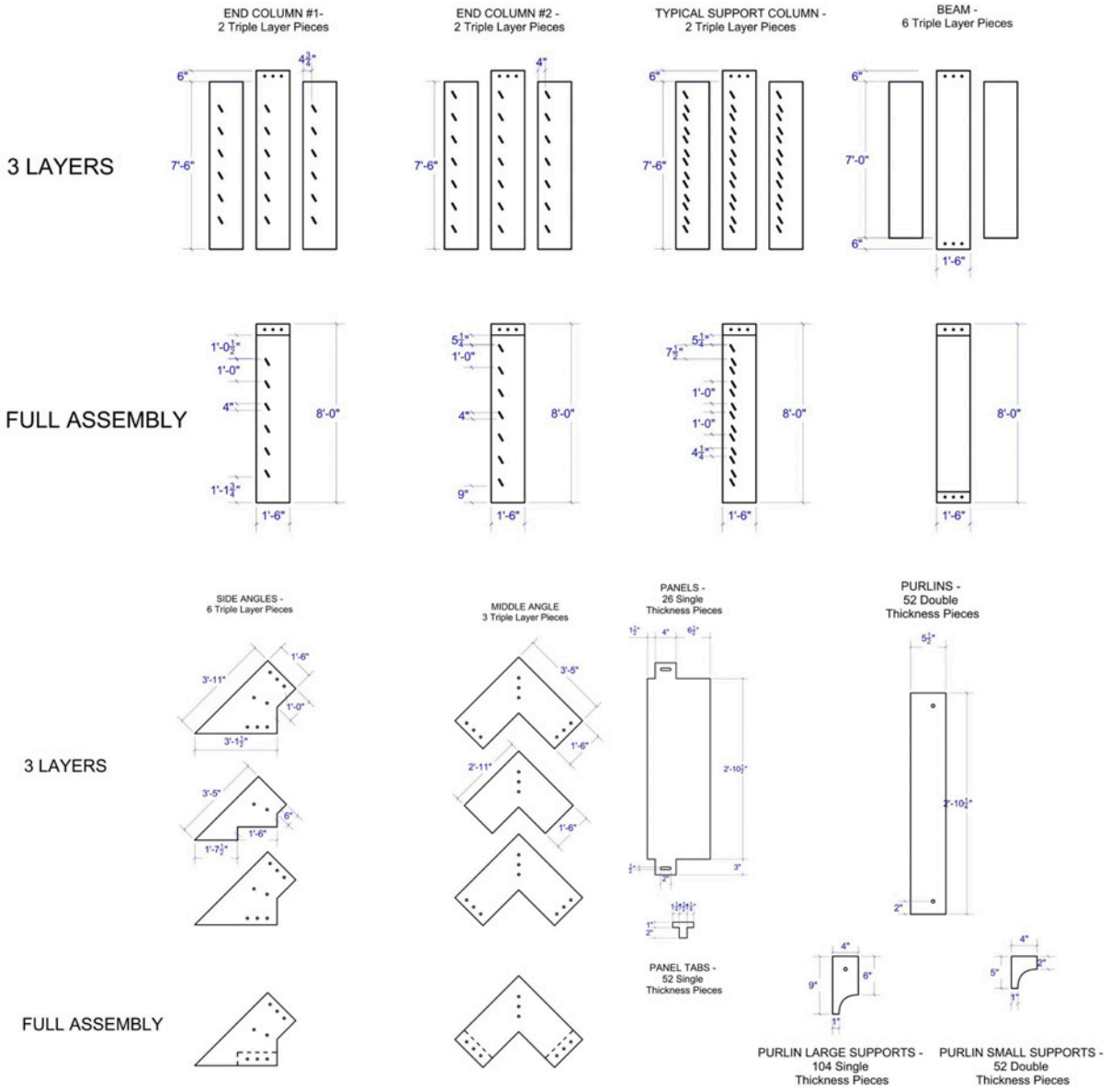
For the constructability review we focused on a large system as we tried to maximize the size of each piece to make sure to use as much of the given 2'x8' panels as possible. This design was built and extended using alternating vertical slats and roof beams, secured to the column and roof frame. The biggest drawback was the initial height of the building being 17 feet. This height was too tall for us to realistically build in the time allowed and was not conducive to the human scale. There was also discussion of needing a simpler system for the roof rafters to allow for Polycarbonate attachments. We also needed to come up with a simpler connection point for the slats placed on the sides for ease of construction.





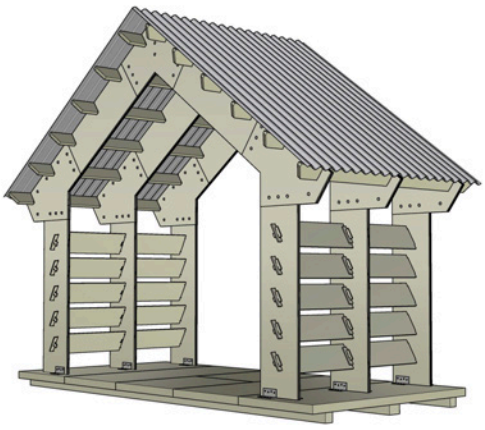
CONSTRUCTABILITY REVIEW
KIT OF PARTS

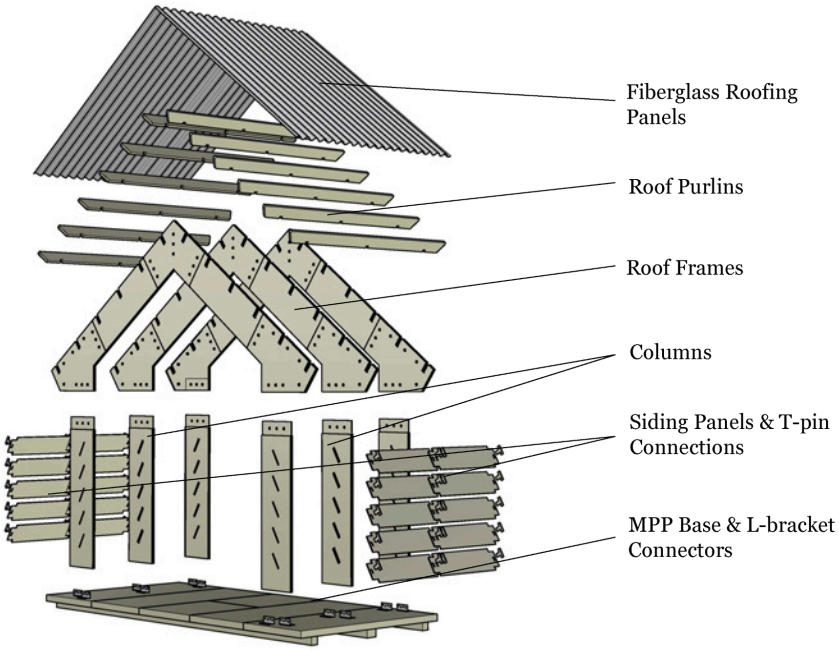
In this kit, there were 23 pieces needed in total to create each of the 11 main components of this kit. This is due to the triple-layering of the beams, columns, and angle pieces. As seen in this kit, there are many small pieces that we realized would have been hard to CNC in order to make the connections in this design.



FINAL KIT OF PARTS

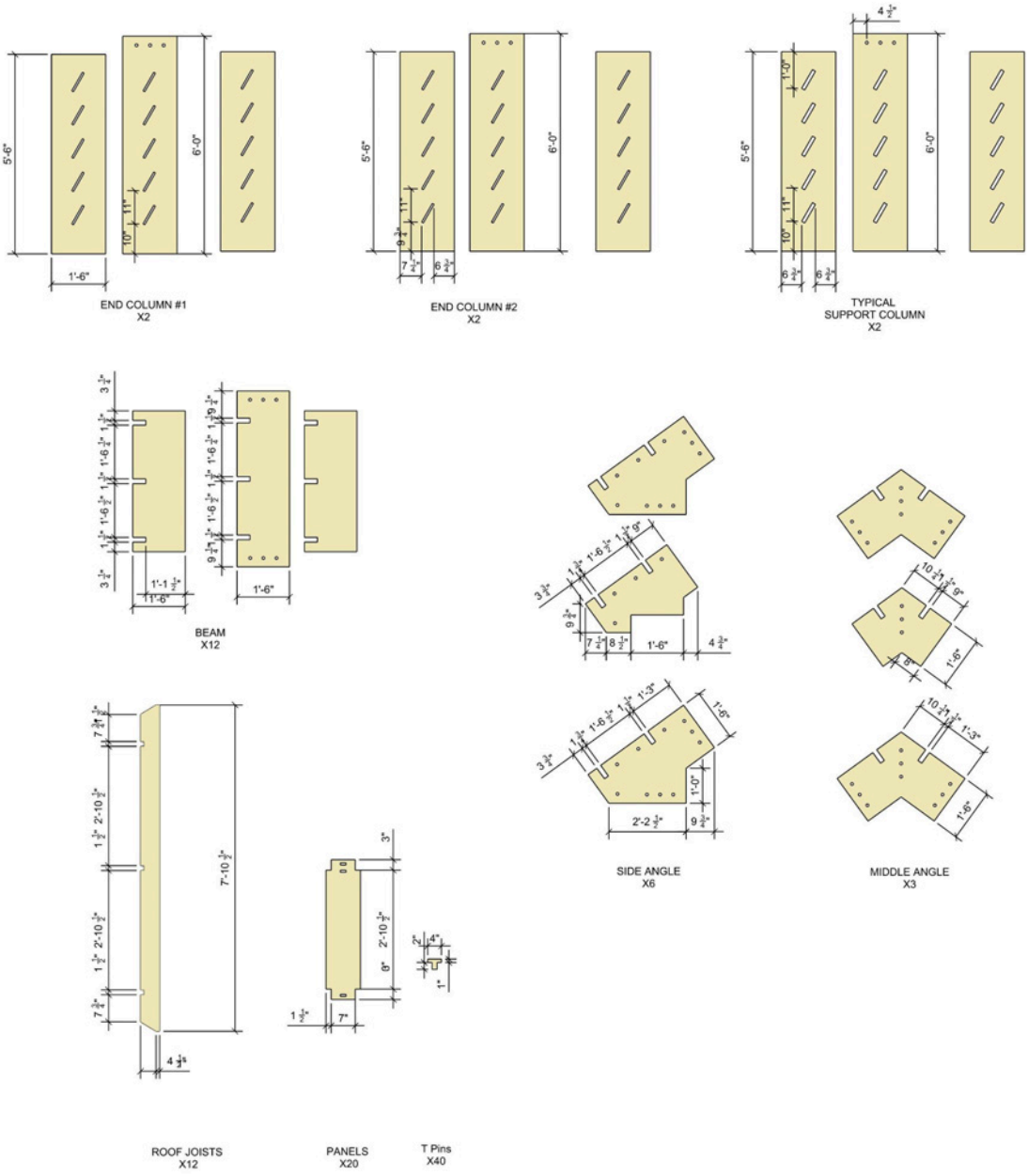
Our final kit of parts focused on the constructability of the structure and was limited to a 12ft. height. The slats were also placed in a way so that the pattern would not appear to alternate from bay to bay, although one layer does sit a ½ inch deeper into the column than the other side so that the slat connections on the middle column could be doubled up. Finally we aimed to gain additional lateral stability with the use of continuous rafters on this prototype, each of them notched three times for stability in their connection to the frame. We also for the sake of constructing this prototype on an unlevel site, built it connected to a MPP base with alternating L-brackets. The fiberglass roofing also became as significant bracing element in this system which we did not anticipate upon designing.





FINAL KIT OF PARTS

In the kit of parts for our final prototype, there are 15 unique pieces and 23 total pieces needed to create one of each of the 9 main components of this kit. This is due to the triple-layering of the beams, columns, and angle pieces. The panels and T-pins consist of single layers of ½ inch plywood and the rafters are made from reclaimed 2x6 lumbar. In comparison to the midterm kit of parts, this kit has less internals cuts, less unique pieces and no small roof purlin pieces to manufacture.

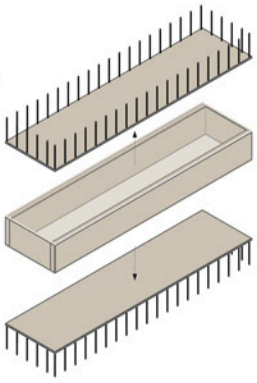


FABRICATION PLAN

Because of the unique challenges with the fabrication and cutting of our triple layer pieces we decided to create a re-manufacturing plan going from the pre-existing panel boxes to our finalized columns and beams. The first step of this process was to disassemble the previous years boxes. We kept the 2x8 side pieces to use as our purlins and took the face panels for CNC cutting. These panels were cut to their appropriate lengths (2 5.5' pieces and 1 6' piece for the columns) and then glued together. Once fully glued and pressed, the next step was to cut the parts using the CNC cut plans. Finally, the columns and beams would be set into the respective mortise joints of the connecting pieces and dowels would be drilled through both pieces. This helped insure that once the structure was assembled, the holes in the columns, beams, and connectors would all line up perfectly.

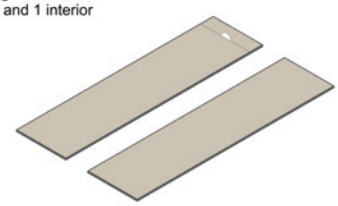
Step 1

Disassemble previous panels by removing nails and staples. Save the 2x6 side pieces to use for purlins and the 2'x8' panels for CNC.



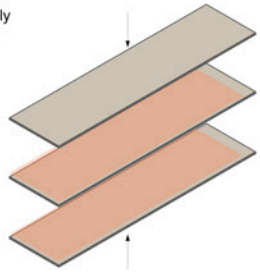
Step 2

Use table saw to cut the length 2 exterior pieces and 1 interior piece.



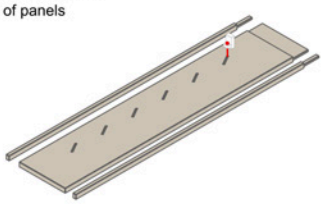
Step 3

Glue assembly together and press. Wait until fully dried to start next step.



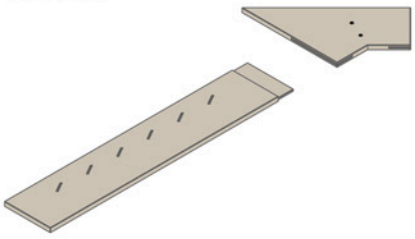
Step 4

CNC shear panel insets using 1/2" diameter drill bit. Cut width of panels



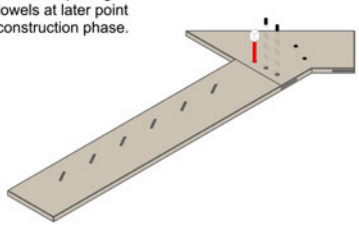
Step 5

Set within angle frame member.



Step 6

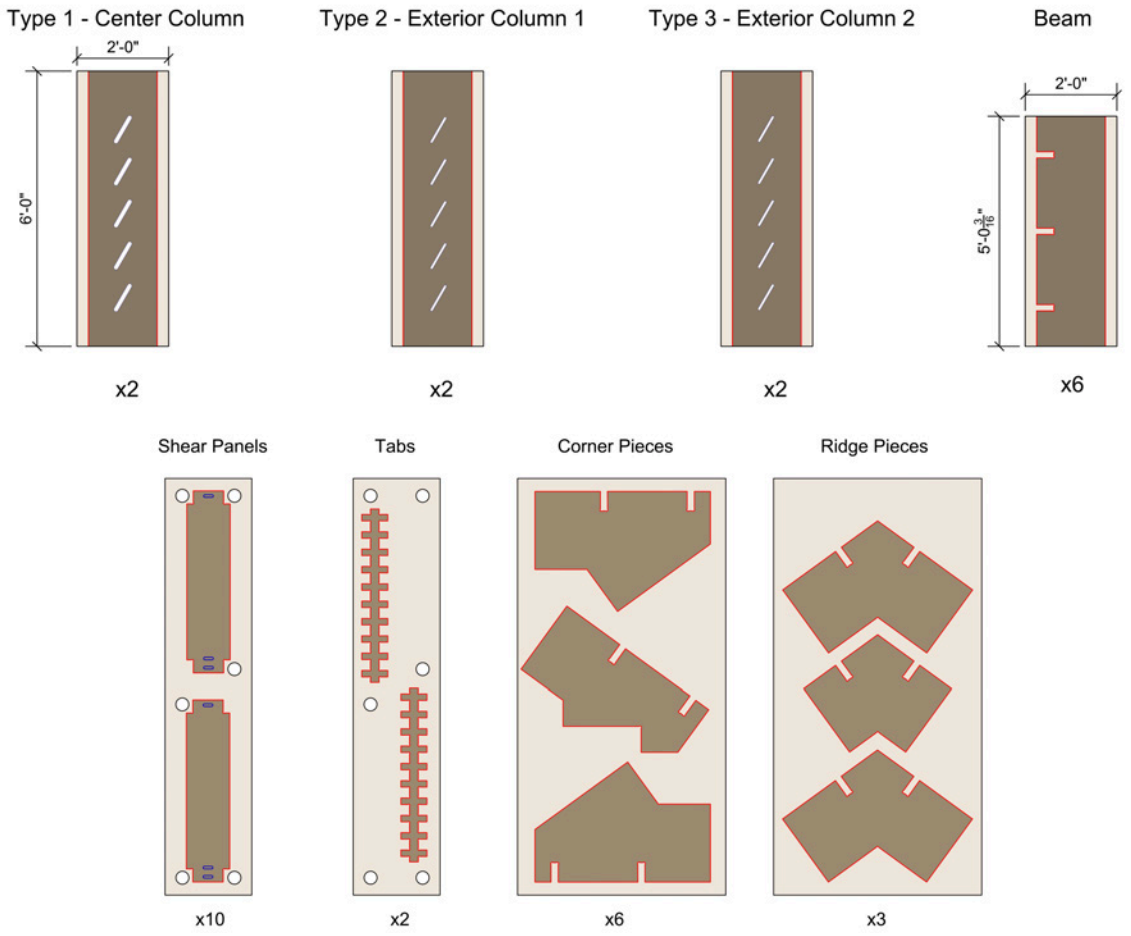
Use drill to create 1" diameter dowel opening. Insert dowels at later point during construction phase.

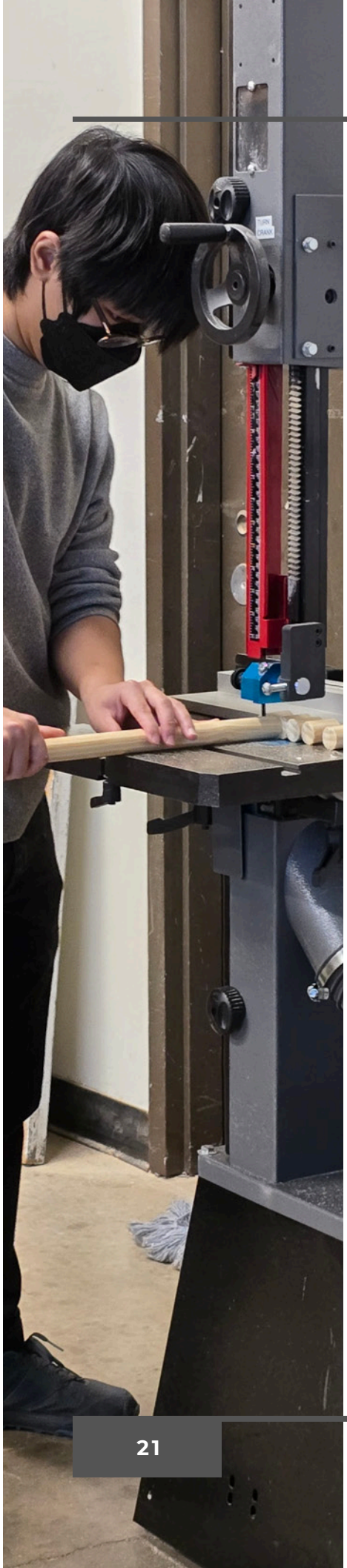


CUT PLANS

The cut plans for the kit of part we're divided up between pieces that had to be cut at OSU and at UO. The pieces that were cut at OSU were developed using our re-manufacturing plan (columns and beams). Because of this, the pieces were already already at the correct length and just had to be CNC cut to trim the edges and add slots for the panel insertions. The original design had to be slightly modified during this step to create curved edges to the inserts for the columns in order to make CNC cutting possible. Additionally, for the beams dog bone cuts were added to interior right angles for the purlin cavities.

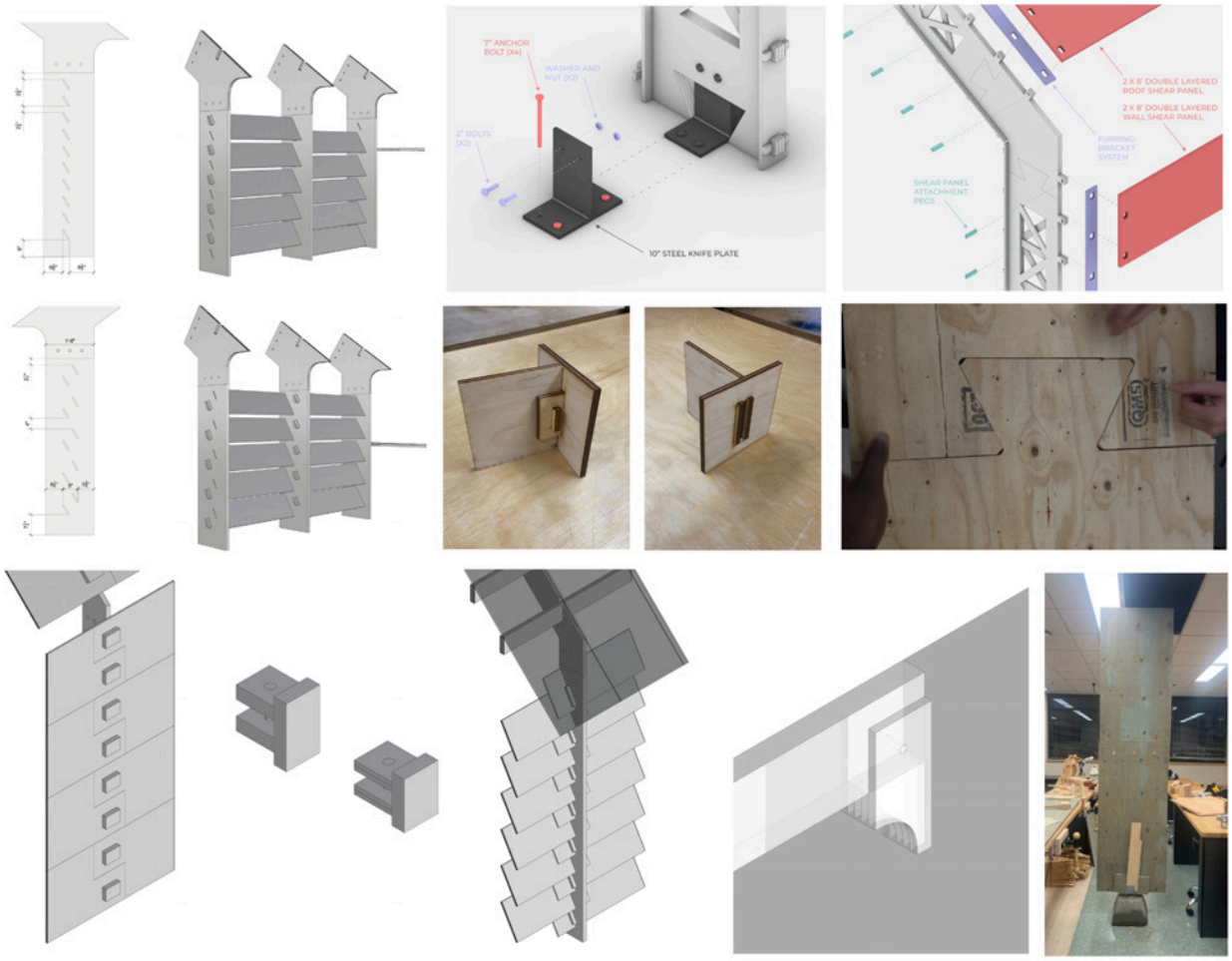
At UO the panels, T-pins, ridge connections, and corner connections were fabricated. Unlike at OSU, the triple layer ridge connections and corner connections could not be creating using our re-manufacturing plan because of the angled nature of the pieces. Because of this, the individual layers were CNC cut then the triple layers were glued and screwed together. These pieces were very simple to layout and the only modification was to create dog bone cuts at interior corners. The single ply T-pins and panels were cut from existing 2'x8' sheets that had pre-existing holes in them. To account for this a master plan for the sheets was developed that accounted for all possible hole locations. After this was done, the pieces were cut out as normal with dog bones added and rounded interior edges in the panels.



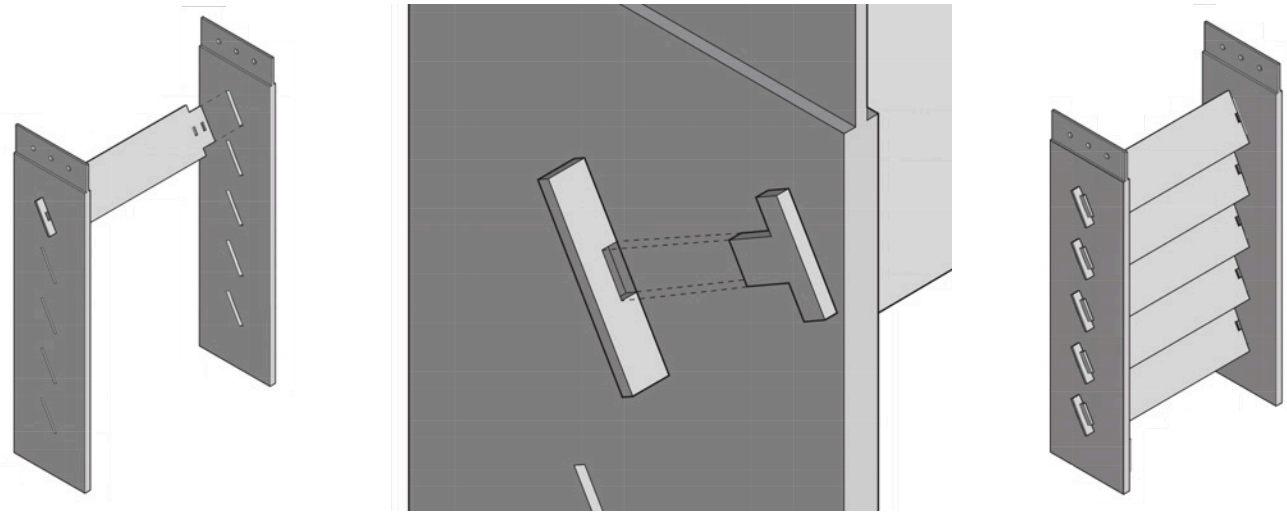


CONNECTIONS

The final connections underwent several iterative phases as we balanced various aspects of design, fabrication, and assembly. Initial concepts were more focused on finding a universal connection type that could work for all types of enclosures and purposes, but this approach was soon discarded in favor of more specialized pieces that were easier to manufacture and assemble. Other connection iterations were disregarded for being too complex to assemble, structurally unstable, or difficult to manufacture using the CNC. The culmination of our iterations were the slats with t-pin, purlin slots, dowel joints, and angle brackets. These connections were chosen for their balance of matching design intent, highlighting wood-to-wood connections, and their ease of assembly and manufacturing.

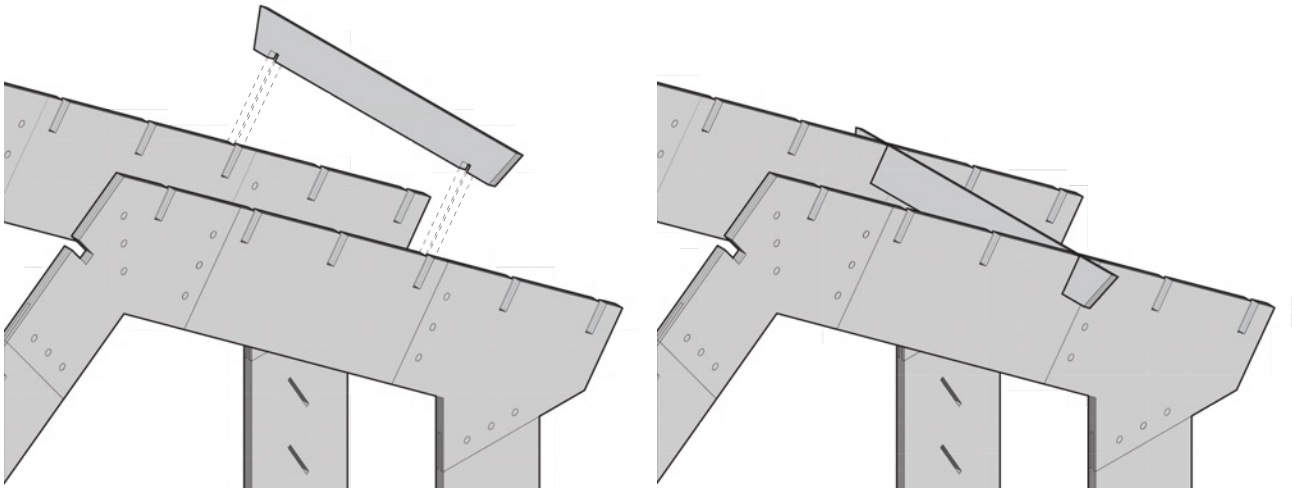


Various scrapped connections developed by connections team and studio

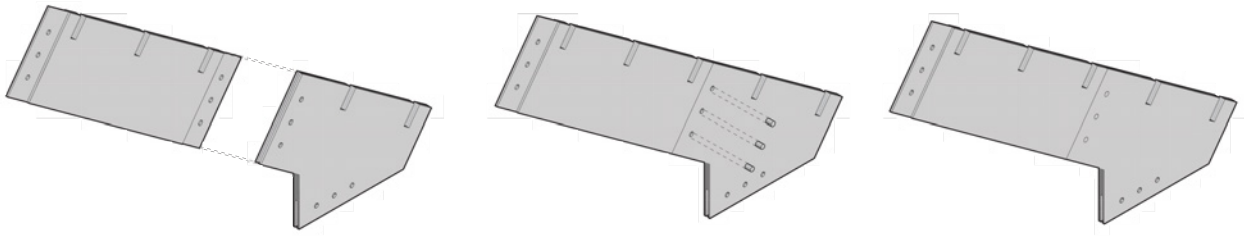


Final slat-column with t-pin connection

FINAL CONNECTIONS

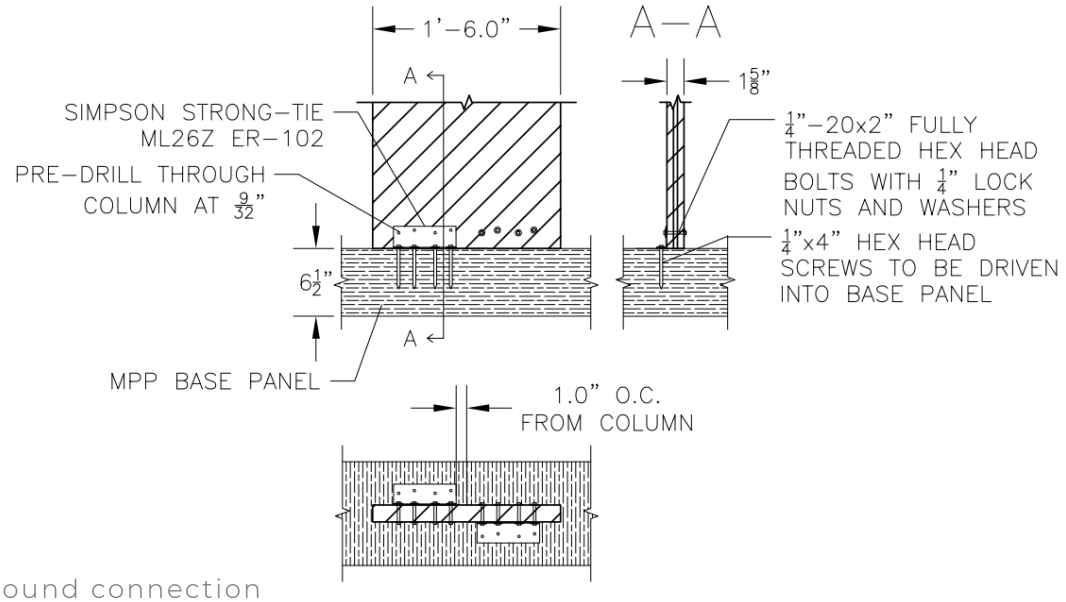


Final rafter-purlin slot connection



Final dowel connection

FINAL CONNECTIONS



Final ground connection

The dowel connection and slat connection received the most feedback during the final review. Regarding the dowel connection, most comments were concerned about why a dowel was chosen instead of a screw and how a dowel may withstand long-term exposure to the elements. This was further supported during the final assembly when screws were still required to secure the connections together. Future projects may want to consider entirely replacing the dowels with screws instead, which provide a quicker assembly that is more secure. Comments over the slat connection largely revolved around the t-pin and its necessity. Upon building the structure, the t-pins were not as structurally significant since the tolerances for openings of the slats were larger, causing the connection to not be as rigid as expected. This caused the t-pins to be possibly entirely removed with little to no effort. Last, for the ground connection, there was a discussion about the accessibility of the structure due to the prototype being built on top of an MPP base panel. To ensure accessibility, investing in a wider concrete footing that remains above the ground instead can support the width of columns while individuals move through the structure.

MOISTURE PROTECTION

Building Enclosure

Key Functions of the Enclosure

- Protect structure from precipitation, wind, and moisture intrusion
- Support ventilation and drying of wood components
- Maintain durability while preserving the lightweight kit-of-parts logic

Primary Strategies

- Layered fiberglass roof assembly for rapid shedding of water
- Vertical siding configuration promoting drainage and airflow
- Material protection coatings to limit water absorption

Foundation Separation

- Elevated Base: MPP panels rest on a 15 x 7 ft panel bottom and foundation MPP surface

Roof Geometry

- A sloped roof encourages rapid runoff away from structural joints
- Overhangs reduce exposure of columns and wall panels

Siding Orientation

- Vertical panel layout promotes downward water movement.
- Open-joint design allows drainage and airflow behind panels

Moisture Protection

- Flex-Seal applied to end-grain areas most vulnerable to capillary uptake
- Clear coatings protect without altering appearance or tolerances

Drying Pathways

- An air cavity behind siding prevents trapped moisture
- Panel spacing ensures evaporation and reduces long-term swelling

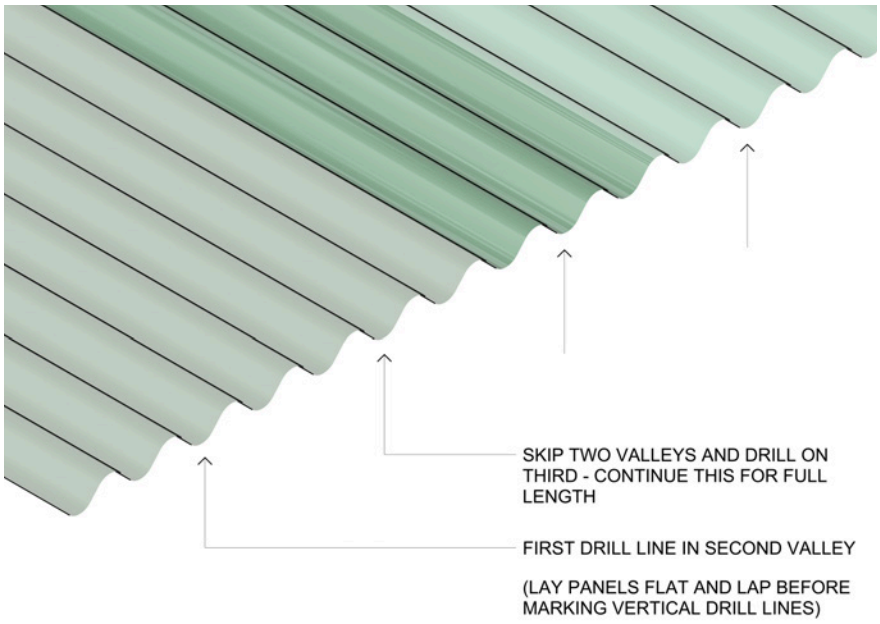
Corrugated Roofing System:

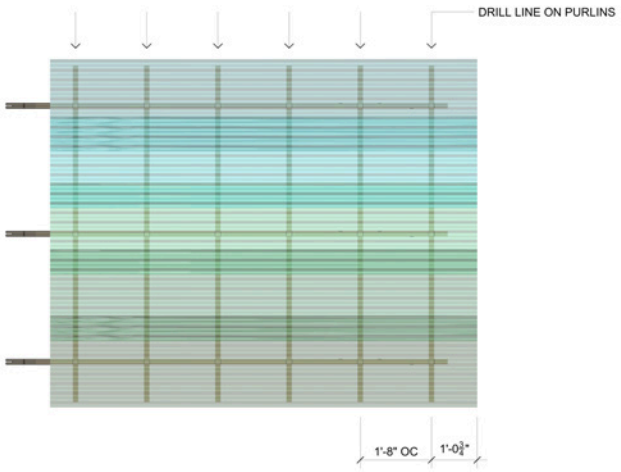
- 20 count 2 x 10" clear fiberglass corrugated roofing panels.
- 2 count ridge caps.
- 200 count washered roofing fasteners.
- Horizontal closure strips (optional).

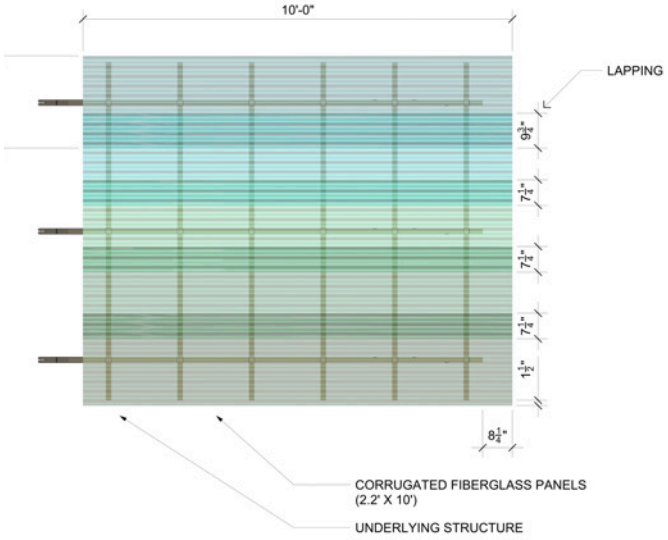
(Graphics below show half of roof profile, colors differentiate individual panels)

Roof Design Considerations:

- Panelized system is lightweight and easy to assemble / disassemble.
- Fiberglass panels are highly resistant to weather.
- Corrugated roofing is readily available.
- Panels easily attach to roof structure.
- Translucent panels reveal wood connections and structure underneath.







1. Mechanisms of Water Transfer in Wood

Water moves through wood by four main mechanisms: **capillarity**, **gravity flow**, **diffusion**, and **convection**.

·**Capillarity**: Liquid water drawn through microscopic pores and cell walls by surface tension.

·**Gravity flow**: Downward movement of liquid water through cracks, checks, or lamination gaps under the influence of gravity.

·**Diffusion**: Water vapor migration through pores, driven by differences in vapor concentration (humidity gradients).

·**Convection**: Water vapor carried by moving air through larger openings or voids.

For the prototype, the most critical mechanisms are capillarity and gravity flow, since the MPP panels were exposed to standing water during testing. Moisture vulnerability was greatest at surface seams, panel edges, and end grain, where gaps and exposed fiber structure allowed faster absorption.

2. Prototype Moisture-Protection Strategies

The prototype utilized a dual-coating system to minimize both surface and end-grain water absorption:

·**LotusPro (clear hydrophobic coating)**: Applied to panel faces. Forms a breathable water-repellent barrier that bonds with cellulose, repelling liquid water while allowing vapor exchange.

·**Flex Seal (clear rubberized coating**): Applied to end grain. Creates a flexible, waterproof seal that blocks capillary water uptake in high-absorption zones.

·**Surface slope detailing**: Added to promote runoff and reduce prolonged water contact.

Together, these treatments significantly reduced capillary absorption and gravity-driven flow, improving overall moisture resistance and dimensional stability.

3. Material Properties and Selection Criteria

Both coatings were selected for their durability, low VOC content, adhesive compatibility, and transparent finish suitable for exposed wood applications.

·**LotusPro**: Nano-silane formulation with high hydrophobicity, UV stability, and long-term adhesion to wood fibers.

·**Flex Seal**: Elastomeric coating forming a continuous, flexible barrier that tolerates dimensional movement and prevents cracking.

Application: Apply to clean, dry surfaces at 15–25 °C, two coats with full drying between applications.

Maintenance: Reapply every 3–5 years or after visible wear, surface damage, or UV degradation.

4. Effects on Construction Tolerances

Both coatings form very thin layers (<0.1 mm) and do not impact construction tolerances or fit.

Edge treatments may require light sanding before gluing to maintain proper bonding.

Overall, the coatings enhance durability and moisture control without affecting structural precision or assembly.

5. Proposed Modifications to the Enclosure

Following prototype testing and final review feedback, several improvements were proposed to enhance performance and durability:

·Larger roof overhangs to minimize direct rainfall exposure and reduce the frequency of wetting-drying cycles.

·Using corrugated roofing material as siding to protect structure. Siding and roofing could be continuous profile.

·Graffiti-resistant finishes or protective coatings on visible surfaces to improve maintainability and appearance.

·Increased clearance at column base connections to prevent water pooling and allow proper drainage and inspection access.

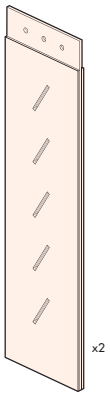
These modifications further strengthen the building's moisture protection strategy and improve long-term serviceability of the timber enclosure.

CONSTRUCTION PLANNING

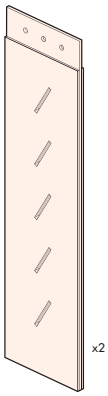
- To prepare the Hatfield Courtyard for our structure, we selected the most flat surface which was towards the southern end of the courtyard. We started by laying down four pieces of solid-sawn lumber (two 4x4x15' members and one 4x6x15' member). Next, we placed six total MPP on top of these members to create a platform for our structure. The sizing of the MPP came out to be four 3'-0"x7'-0" and two 3'-0"x3'-6" panels. We connected these panels together with 32 screws total, screwing diagonally to ensure a strong connection
- Our assembly process was split into four major categories, being **Prefabrication**, **Transportation**, **Site Preparation**, and **Construction**. We prefabricated the walls and frame tops of the structures, carrying them to site by hand using 3-4 people, assembled our foundation platform, then constructed the rest of the structure on site. Our process is further documented with diagrams.

CONSTRUCTION SEQUENCING - KIT OF PARTS

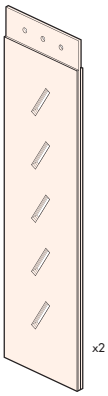
Fabricated Timber Elements


x2

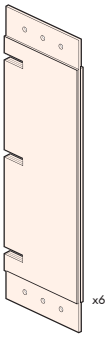
Typical End Column 1


x2

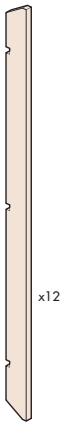
Typical End Column 2


x2

Typical Middle Column


x6

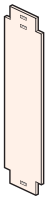
Typical Roof Beam


x12

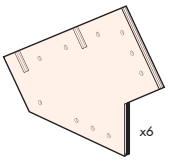
Typical Roof Purlin
(made from 2x6)

x40

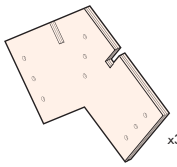
T-Pin


x20

Wall Panel

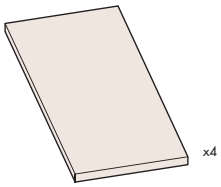

x6

Side Angle Module

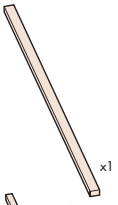

x3

Top Angle Module

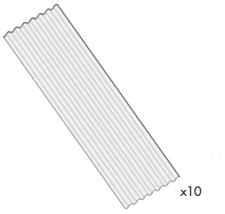
Site, Products, and Tools


x4

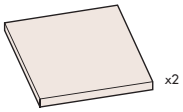
3'-0" x 7'-0" MPP


x1

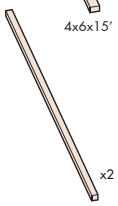
4x6x15'


x10

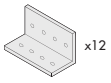
Fiberglass Roof Enclosure


x2

3'-0" x 3'-6" MPP


x2

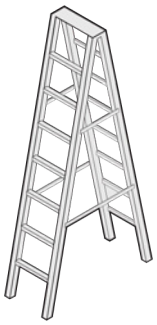
4x4x15


x12

L-Brackets

x2

3'-0" Polycarb Ridge Cap

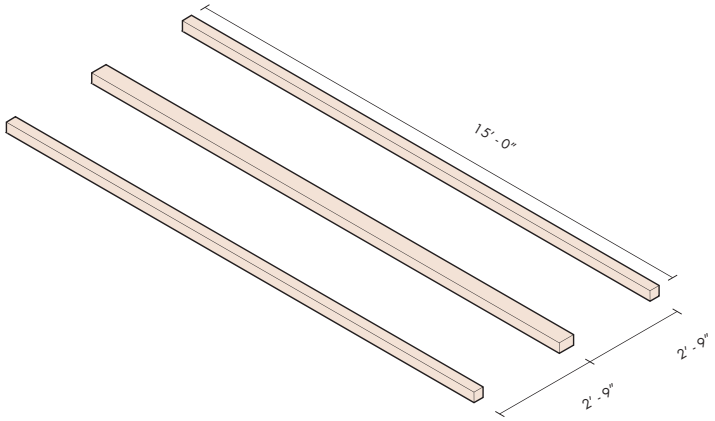


12' Tall Adjustable A-Frame Ladder

Impact Driver

Rubber Mallet

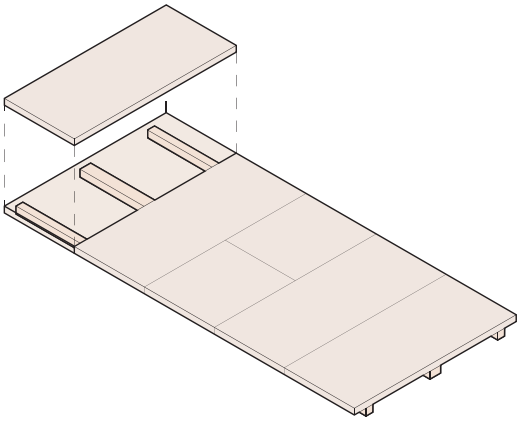
SITE SELECTION AND PREPARATION



x2-4

Lay the two 4x4s on either end of the 4x6 by 2'-9"

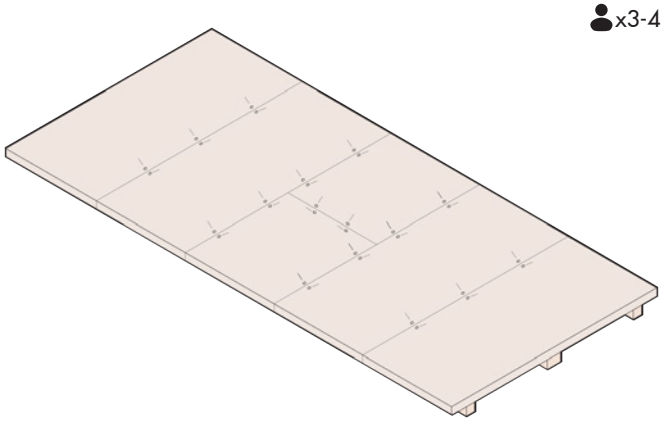
1.



x3-4

Lay the four 3'x7' MPP on top of the 4xs with the two 3'x3.5' MPP in the center

2.



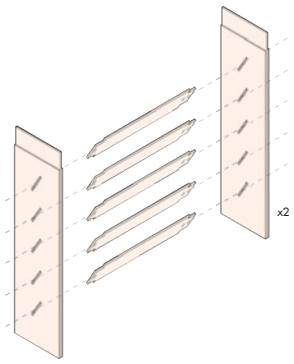
x1-4

With 32 screws, screw diagonaly into the MPP to secure each panel to one another. Additional people may participate in this step to speed up the process.

3.

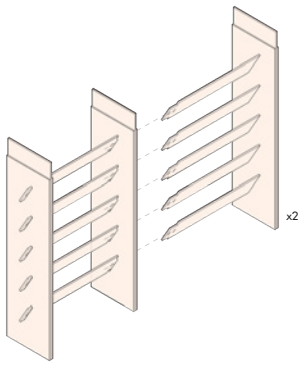
PREFABRICATION

1.



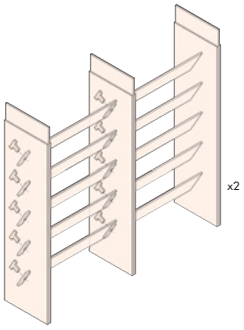
x3-4 Slide five wall panels into a typical edge column 1 and a typical middle column. Two people should hold the columns in place while another slides the panels into place

2.



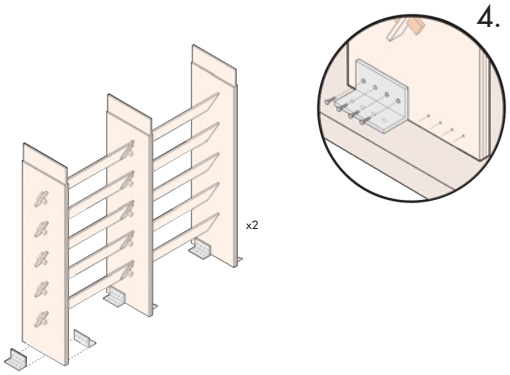
x3-4 Slide a wall panel connected to a typical edge column 2 into a completed partial wall from step 1

3.



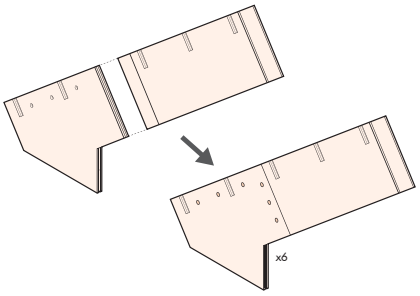
x1-2 Slide twenty T-pins into each of the wall panel holes to secure them in place

4.



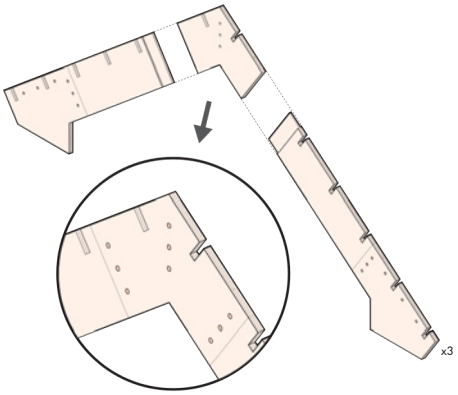
x2-3 Slide twenty T-pins into each of the wall panel holes to secure them in place. 1-2 People should hold the walls in place while a third bolts the brackets in

5.



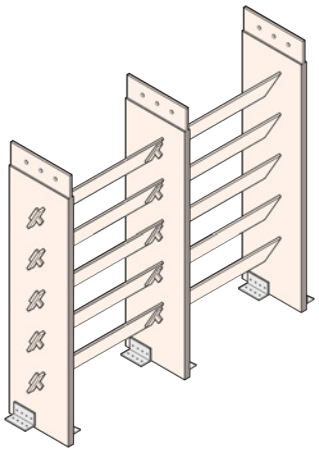
x3-4 Slide a typical column into a side angle module.
Drill 1" holes once connected and secure with dowels hammered into place

6.

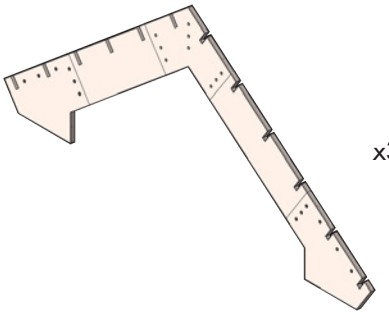


x3-4 Slide two modules constructed from step 5 into a top angle module.
Drill 1" holes once connected and secure with dowels hammered into place

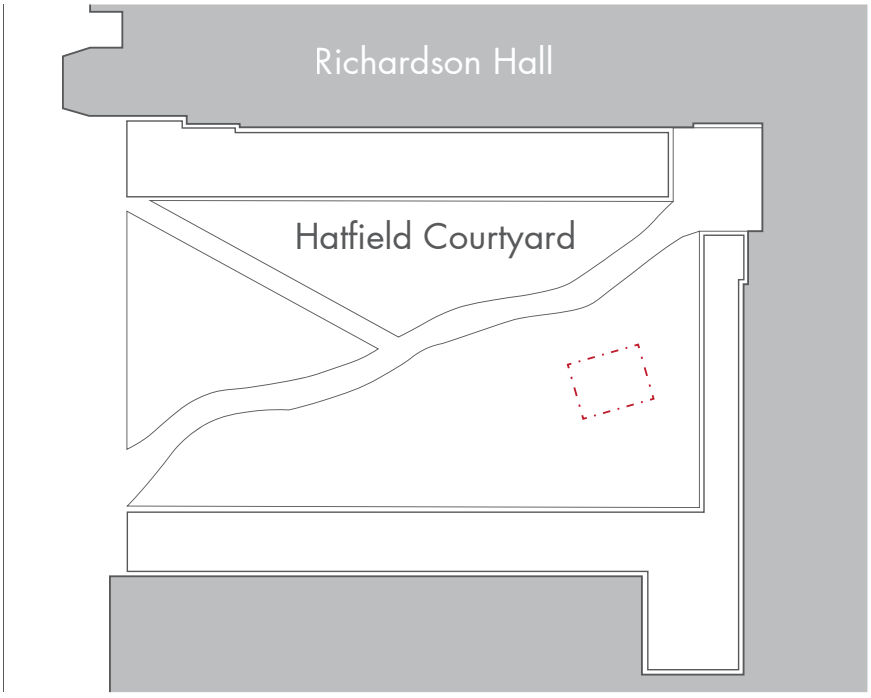
TRANSPORTATION



x2



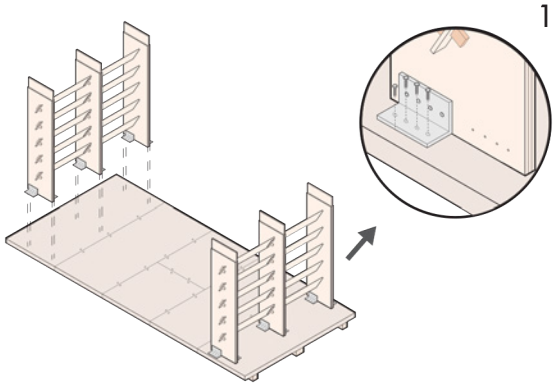
x3



Transport the prefabricated walls and frame tops to site. Each prefabricated element can be carried by 3-4 people

ON-SITE CONSTRUCTION

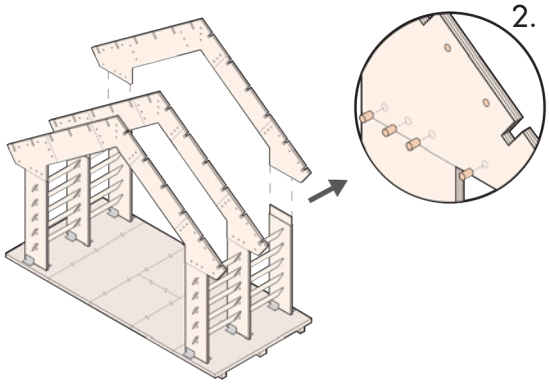
1.



x3-4

Place the prefabricated wall modules centered on the MPP foundation.
Secure each column with eight bolts through the L-Brackets

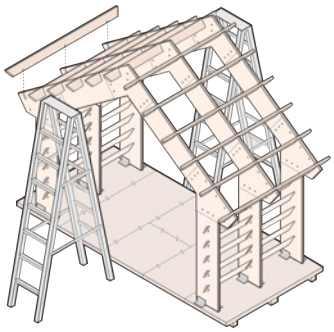
2.

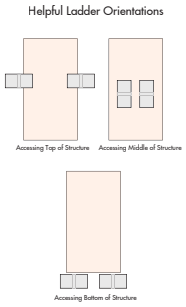


x4-5

Place each of the frame tops on the columns.
Drill 1" holes once connected and secure with dowels hammered into place

3.

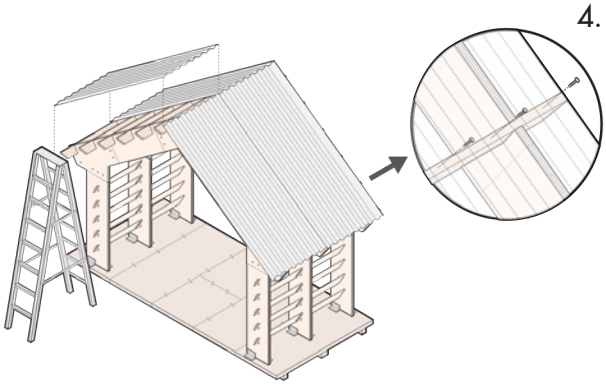




x3-4

Hammer in place the twelve 2x6 purlins into the gaps in the frames.
Use A-Frame Ladders and locations highlighted to achieve the different heights of the structure

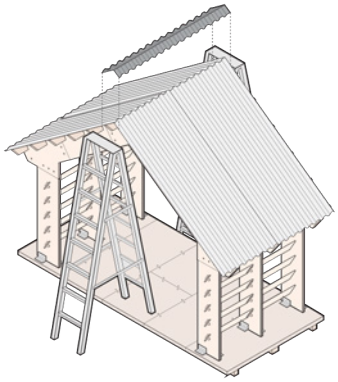
4.



x3-4

Screw Fiberglass roof panels into the purlins at each "hill" of the fiberglass.
Use A-Frame Ladders and locations highlighted in step 13 to achieve the different heights of the structure

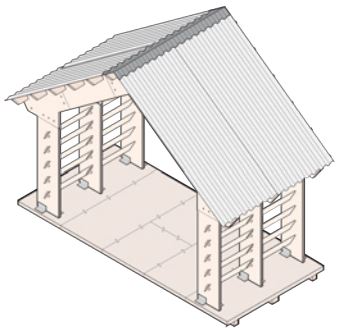
5.



x2-3

Screw ridge cap through the fiberglass into the tops of the frames.
Use A-Frame Ladders and location highlighted in step 13 to access the top of the structure

6.



RISK MANAGEMENT PLAN

SAFETY IN DESIGN, CONSTRUCTION PLANNING, AND SITE

- **Design**

Informing design decisions based on safety and lowering the amount of risk was the start to the safety planning. The initial design was significantly taller and due to the risk of working from taller ladders, the design was changed to bring the height to 12' rather than 16'.

- **Construction Planning**

Creating a safe and viable construction plan was crucial in making the process safer for both the construction team and the OSU community who would pass by. All lifting and construction operations were planned to account for safe lifting procedures and keeping any danger contained within the building area. No parts above 25 lbs. were to be lifted by any individual and team lifting was implemented throughout the construction plan.

- **Site**

The entire construction area was roped off with signage attached to prevent unauthorized individuals from entering the construction space. This keeps the both the public safer during construction and the team safer from distractions.

CONSTRUCTION PHASES, HAZARDS, LIKELIHOOD, SEVERITY & MITIGATION

Phase 1 – Site Prep & Materials Handling

Hazard	Likelihood	Severity	Mitigation
Dropping material	Medium	Moderate	Wear gloves and boots, proper lifting technique, team lifting for heavy items
Trip hazards	Medium	Low	Material staging area, maintain clean work zone
Splinters & sharp edges	Medium	Low	Gloves, tool awareness

Phase 2 – Prefabrication and Drill Use

Hazard	Likelihood	Severity	Mitigation
Drill injury to wrist/fingers	Medium	Moderate	Training in tool handling & torque control
Wood chips to eyes	High	Low–Moderate	Mandatory eye protection
Incorrect fastening/ misalignment	Medium	Low	Work done at stable stance, verify alignment

Phase 3 – Ladder Use (Roof Installation & Upper Fastening)

Hazard	Likelihood	Severity	Mitigation
Dropping material	Medium	Moderate	Wear gloves and boots, proper lifting technique, team lifting for heavy items
Trip hazards	Medium	Low	Material staging area, maintain clean work zone
Splinters & sharp edges	Medium	Low	Gloves, tool awareness

Phase 4 – Fiberglass Roof Installation

Hazard	Likelihood	Severity	Mitigation
Sheet catching wind	Low	High	Install in low-wind conditions
Loss of footing	Low	Moderate	Stable ladder position, trained users only
Edge hazard handling	Medium	Low	Gloves to protect from sharp fiberglass edges
Inhalation of fiberglass dust	Low	High	N95 respirators for lung protection

TRAINING, PPE, INCIDENTS, COMPLIANCE

Training Requirements

- Ladder-safety training required for all individuals working above 6’
- Tool-use training for drill handling
- Fall hazard awareness training for ladder operators
- Only trained personnel permitted to perform roofing installation

PPE Requirements

- Safety glasses during all fastening operations
- Gloves during fiberglass and lumber handling
- Closed toe construction-grade footwear encouraged

Emergency and Incident Preparedness

- First Aid kit available nearby
- One individual designated as Safety Coordinator
- Immediate reporting of near-miss and incidents

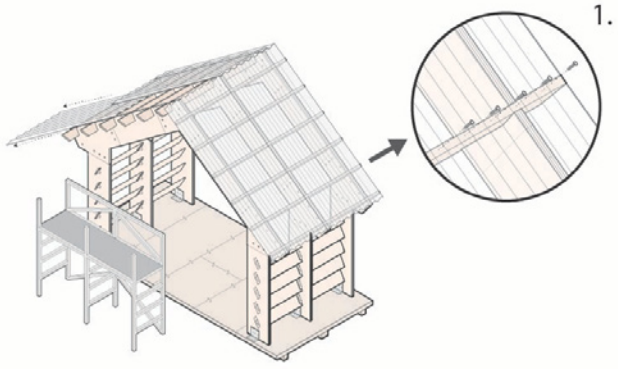
Compliance and Verification

Daily checklist:

- Ladder inspection
- Drill condition and battery integrity
- Work-area housekeeping
- PPE verification

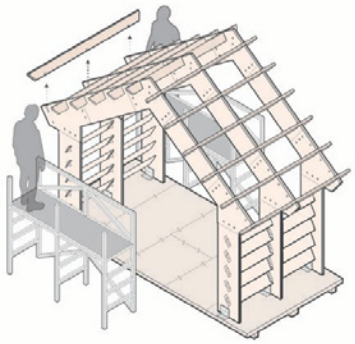
DECONSTRUCTION PLAN

1.



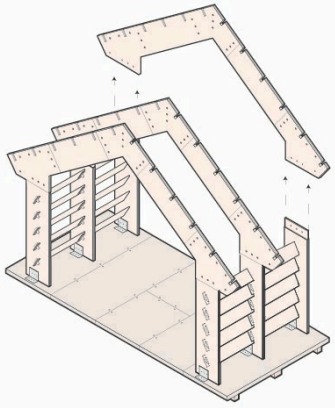
Unscrew fiberglass panels from the purlins
3 people per panel, 1 drill per panel, 1 scaffolding or ladder

2.



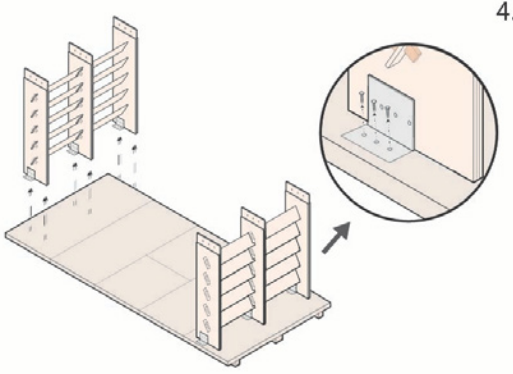
Hammer the purlins out of place, take them out and handoff to someone on the ground
3 people, 2 rubber mallets, 2 scaffoling or ladders

3.



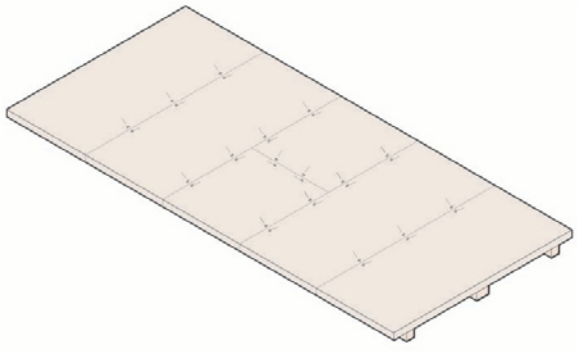
Remove the dowels from the corner joints and remove the top piece as a whole by tilting it down
4-8 people, 2 rubber mallets, 1 scaffolding or ladder per frame

4.



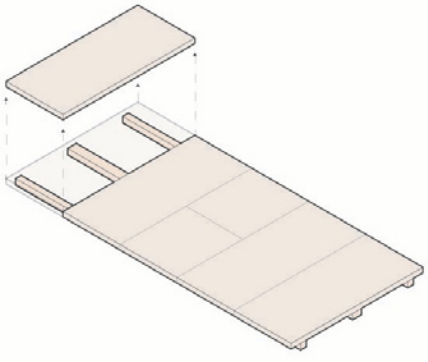
Remove the screws from bracets and remove the walls from the foundation as a whole piece
3 people per wall, 1 drill per wall

5.



Remove the screws from the MPP panels, releasing them from the beams
2-3 people, 2-3 drills

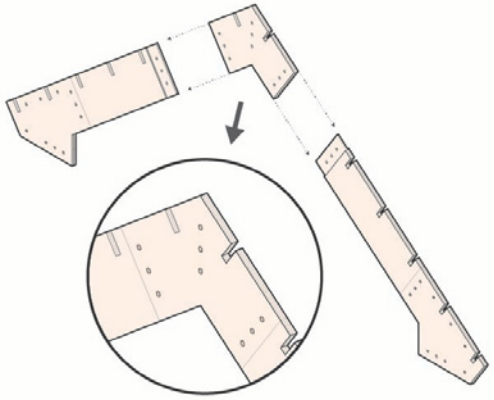
6.



Remove the MPP panels from the beams
4-6 people per panel

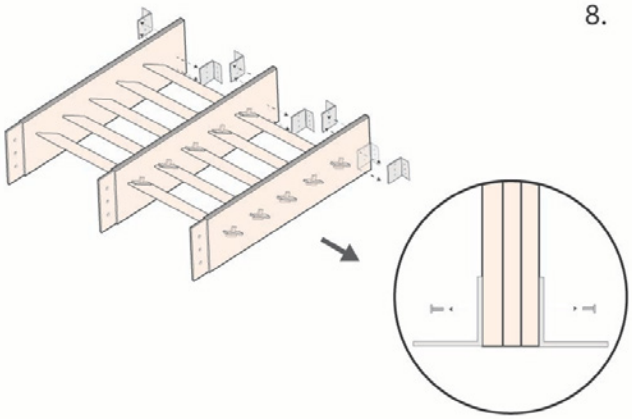
DECONSTRUCTION PLAN

7.



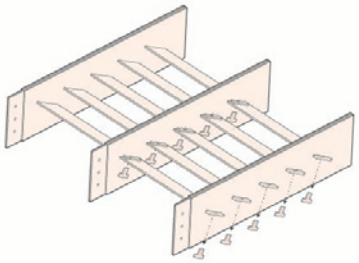
Using rubber mallets, knock out the dowels holding the pieces together and deconstruct the arch while on the ground
2 people, 2 rubber mallets

8.



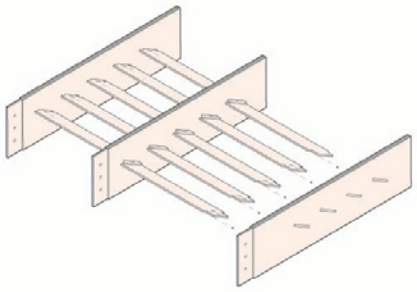
Lay the wall on the side, remove the screws holding the brackets on the columns, then remove the brackets
2 people, 2 drills

9.



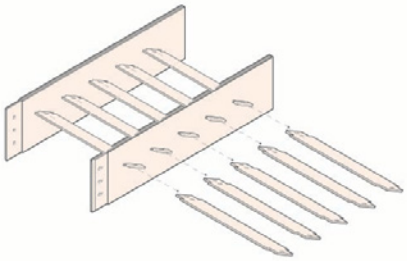
While the wall is in teh side, remove the T-Pins
1 person

10.



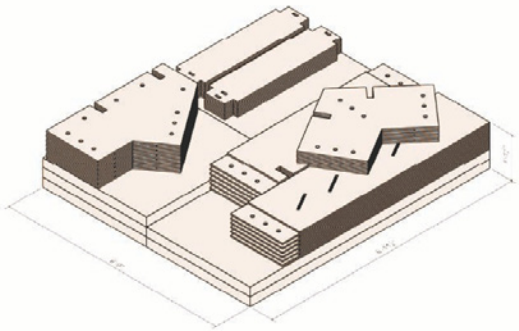
Slide the column off of the panels
2 people

11.



Slide the panels off the the column
2 people

12.



Stack the pieces for storage

POST CONSTRUCTION REFLECTION

SUCCESSES:

- Getting the final projects fully built
- No injuries on site
- Production of pieces completed on time

IMPROVEMENTS:

- Find a more efficient fastener rather than dowels
  - Screwing the joints together would be quicker and safer for the construction
- Using a boom lift
  - Instead of having people on ladders holding up the top piece, using a boom lift to get people out of the way could improve safety
- Redesigning wall slats to allow for tilting up the frame as a whole

REFLECTION

Overall, the construction process was very efficient, constructing the whole structure in around 5 hours, and no injuries were sustained, which is a great accomplishment. During our construction, we may have had a couple of hiccups with pieces, but we quickly problem-solved a solution so we could get it done in time. We had the perfect number of people on site for the process. Around 12 people were the ideal amount for the full construction, with 6 people for the addition of the roofing.

MANAGEMENT

PROJECT TIMELINE

WEEK 8

11/17 — Receive feedback on feasibility, finalize design

11/19 — Iterate solutions based on feedback, start CNC

11/21 — Finalize design, finish CNC (UO)

WEEK 9

11/28 — Thanks giving Break

11/26 — Continue sealing pieces

11/24 — Seal CNCed pieces

WEEK 10

12/01 — Finish sealing, preassemble parts and prepare for transport

12/03 — Assemble structure on site & prepare for presentation

12/05 — Final presentation

MATERIAL USAGE & WASTE

RECYCLED WOOD USAGE:

Recycled 2’ x 8’ plywood panels used: **48**

New 4’ x 8’ plywood panels used: **21**

Recycled 2” x 6” beams used: **12**

Total recycled material percentage: 59%

WOOD WASTE:

Wasted plywood per panel: **9 ft²**

Total wasted plywood: **549 ft²**

Total plywood used: **1,440 ft²**

Total wasted material percentage: 38%

FINANCIAL SUMMARY

Consumable Items

Category	Item	Vendor	Item Cost	Quantity	Total Cost	Status:	For Who?
Materials	1/8" plywood x 18" x	Duck Store	$ 22.00	8	$ 176.00	NYC paid	UO
Materials	MDF 3/4" x 48" x 96"	Crosscut Hardwoods	$ 44.65	2	$ 89.30	SCYP paid	UO
Materials	Assorted	Home Depot - Jordan	$ 21.70	1	$ 21.70	SCYP paid	UO
Materials	Plywood CDX	Jerry's Eugene	$ 22.50	8	$ 135.00	SCYP paid	UO
Materials	Shaper Tape	online	$ 22.00	5	$ 110.00	SCYP paid	UO
Materials	Plywood CDX 1/2*4ft	https://www.homedepot.com/pep/	$ 22.90	8	$ 183.20	SCYP Paid	UO
Tools & Mate	Flex Seal Liquid Clea	https://www.homedepot.com/p/FLI	$ 34.98	2	$ 69.96	SCYP paid	UO
Materials	Corrugated Fiberglas	https://www.betterheadforjerrys.co	$ 23.75	10	$ 237.50		OSU
Tools & Mate	4in. Polycarbonate Ri	https://www.betterheadforjerrys.co	$ 19.99	2	$ 39.98		OSU
Tools & Mate	Wood dowels 1 inch*	https://www.homedepot.com/p/Pir	$ 5.48	3	$ 16.44	SCYP paid	OSU
Materials	Plywood CDX 1/2*4ft	https://www.homedepot.com/pep/	$ 22.90	12	$ 323.76	SCYP Paid	OSU
Tools & Mate	Flex Seal Liquid Clea	https://www.homedepot.com/p/FLI	$ 34.98	2	$ 69.96	SCYP paid	OSU
Tools & Mate	Hex Head Wood Scre	https://www.homedepot.com/p/Wo	$ 6.98	2	$ 13.96	SCYP paid	OSU
Tools & Mate	Wood dowels 1/4" * 4	Home Depot	$ 2.08	1	$ 2.08	Matt paid	OSU
Tools & Mate	Wood glue (one gallo	Home Depot	$ 38.98	1	$ 38.98	Matt paid	OSU
Tools & Mate	Assorted Sandpaper	Home Depot	$ 21.96	1	$ 21.96	Matt paid	OSU
Tools & Mate	Clear Wood Sealer	Home Depot	$ 92.98	1	$ 92.98	SCYP paid	UO
				Total	$1,642.76		

Reusable Item

Category	Item	Vendor	Item Cost	Quantity	Total Cost	Status:	For Who?
Tools & Mate	Router bits 1/4" flat e	https://www.amazon.com/SpeTool-	$ 61.99	1	$ 61.99	SCYP paid	UO
Tools & Mate	Router bits tapered b	https://www.amazon.com/Foos-Tap	$ 39.99	1	$ 39.99	SCYP paid	UO
Tools & Mate	Router bits asst'd bal	https://www.amazon.com/FOOS-Sp	$ 29.99	1	$ 29.99	SCYP paid	UO
Tools & Mate	Hex Head Screw Bolt	https://www.amazon.com/Stainless	$ 10.48	3	$ 31.44	Mariapaola	OSU
Tools & Mate	Nuts & Washers	ACE Hardware	$ 16.25	1	$ 16.25	Luca paid	OSU
Tools & Mate	Sample Nuts, Bolts &	ACE Hardware	$ 28.43	1	$ 28.43	Toraja paid	OSU
Tools & Mate	4 in. x 3/8 in. Shed Re	Home Depot	$ 5.77	4	$ 23.08	SCYP paid	UO
Tools & Mate	Paint rollers	Home Depot	$ 22.68	1	$ 22.68	Matt paid	OSU
				Total	$ 253.85		

Vehicle Rentals

Category	Item	Vendor	Item Cost	Quantity	Total Cost	Status:	For Who?
Transportati	Minivan Rental Oct 3	UO Motorpool	73 + gas	1	$ 172.52	SCYP paid	UO
Transportati	Sedan Rental Oct 3 `	UO Motorpool	39 + gas	1	$ 115.75	SCYP paid	UO
Transportati	Home Depot Truck Re	Home Depot	$ 21.85	1	$ 21.85		UO
				Total	$310.12		

AMOUNT AVAILABLE (SCYP + STUDIO FEES):

$2,400

AMOUNT SPENT::

$2,206.73

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ACKNOWLEDGEMENTS

Special thanks to Tom Coates, Phil Mann, Byrne Miyamoto, Art Myers, Sustainable City Year, the City of Tualatin, Arxada, and all the folks at UO and OSU who made this project possible.

REFERENCES:

- NDS 2024